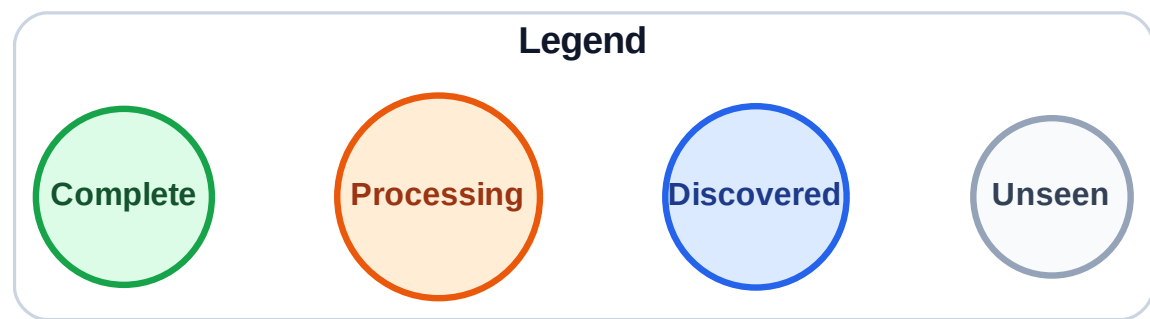
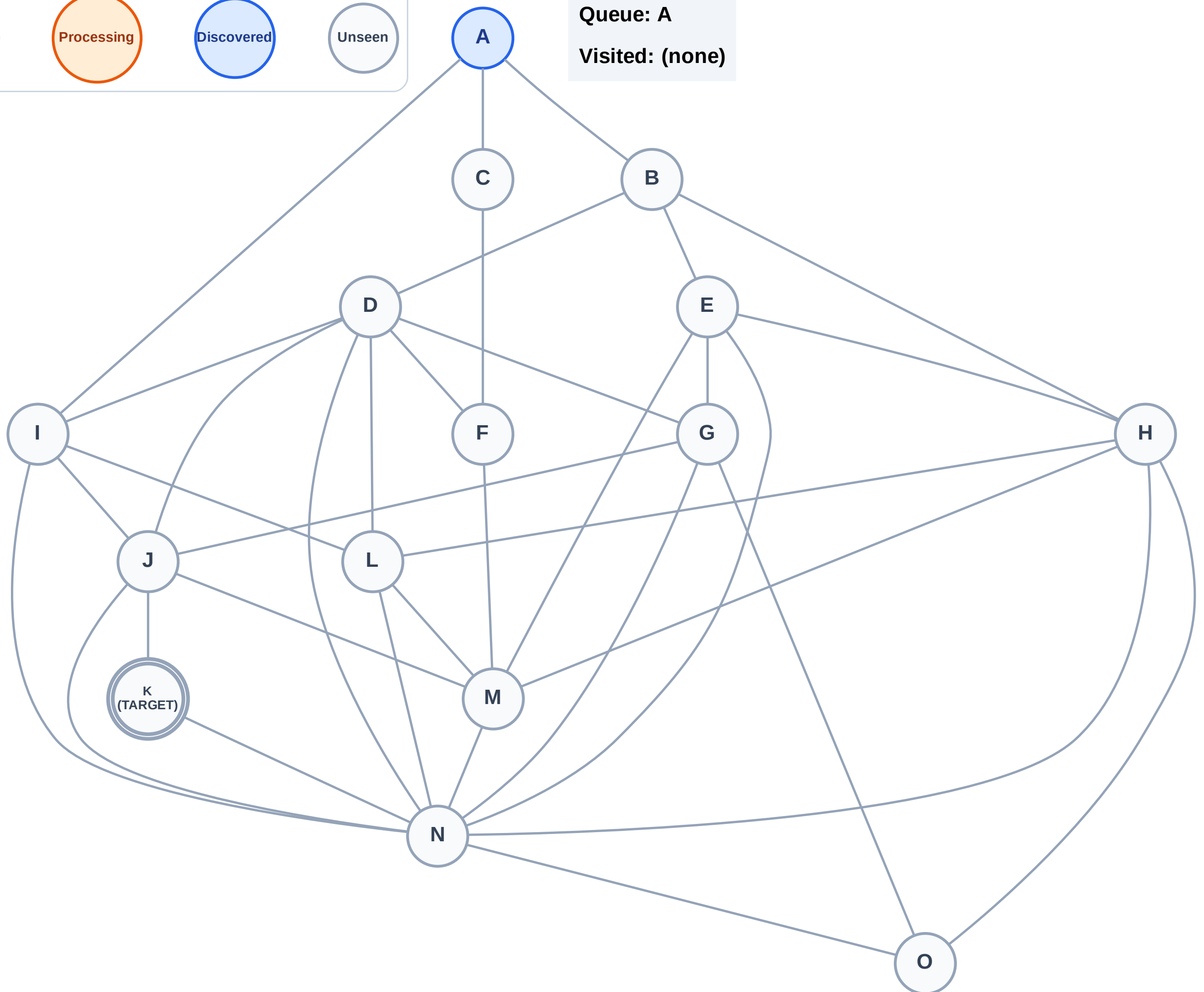


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



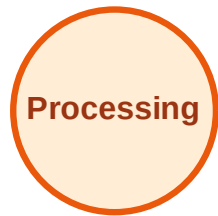
Queue: A
Visited: (none)



BFS Traversal

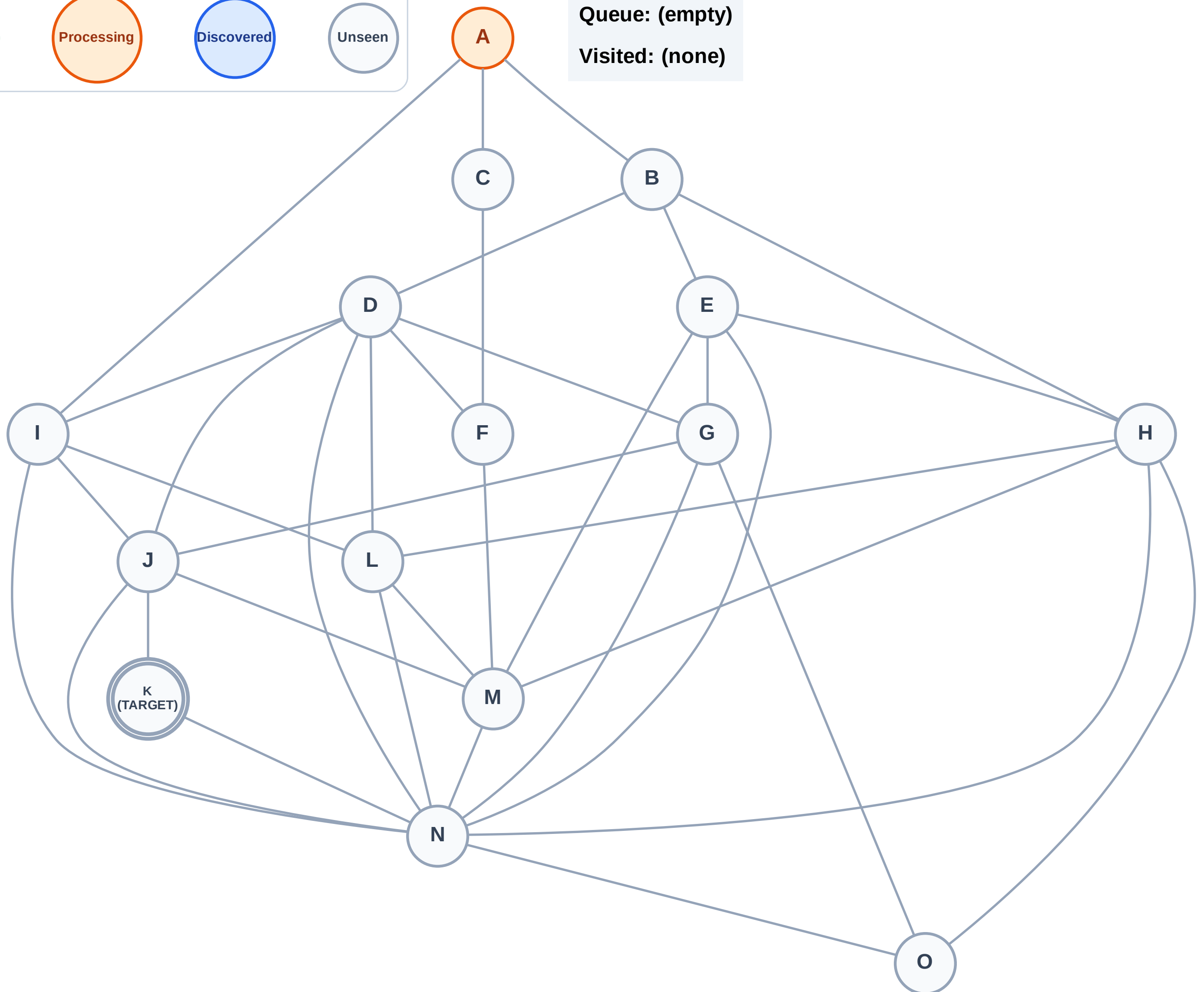
Step 2: Process node A

Legend



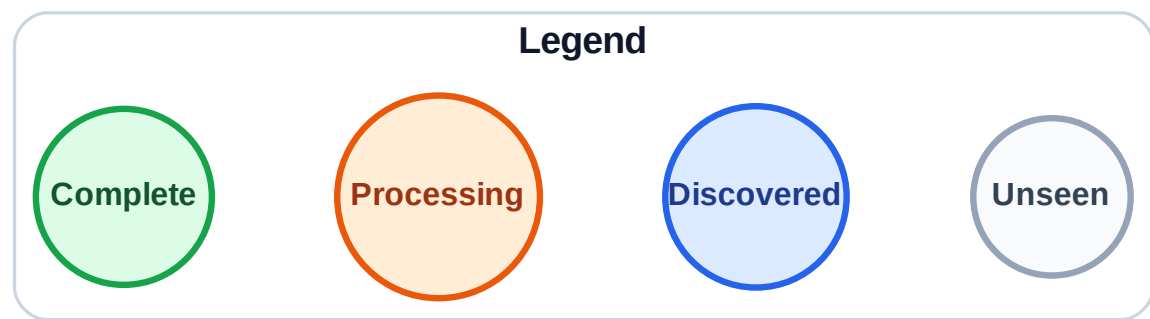
Queue: (empty)

Visited: (none)



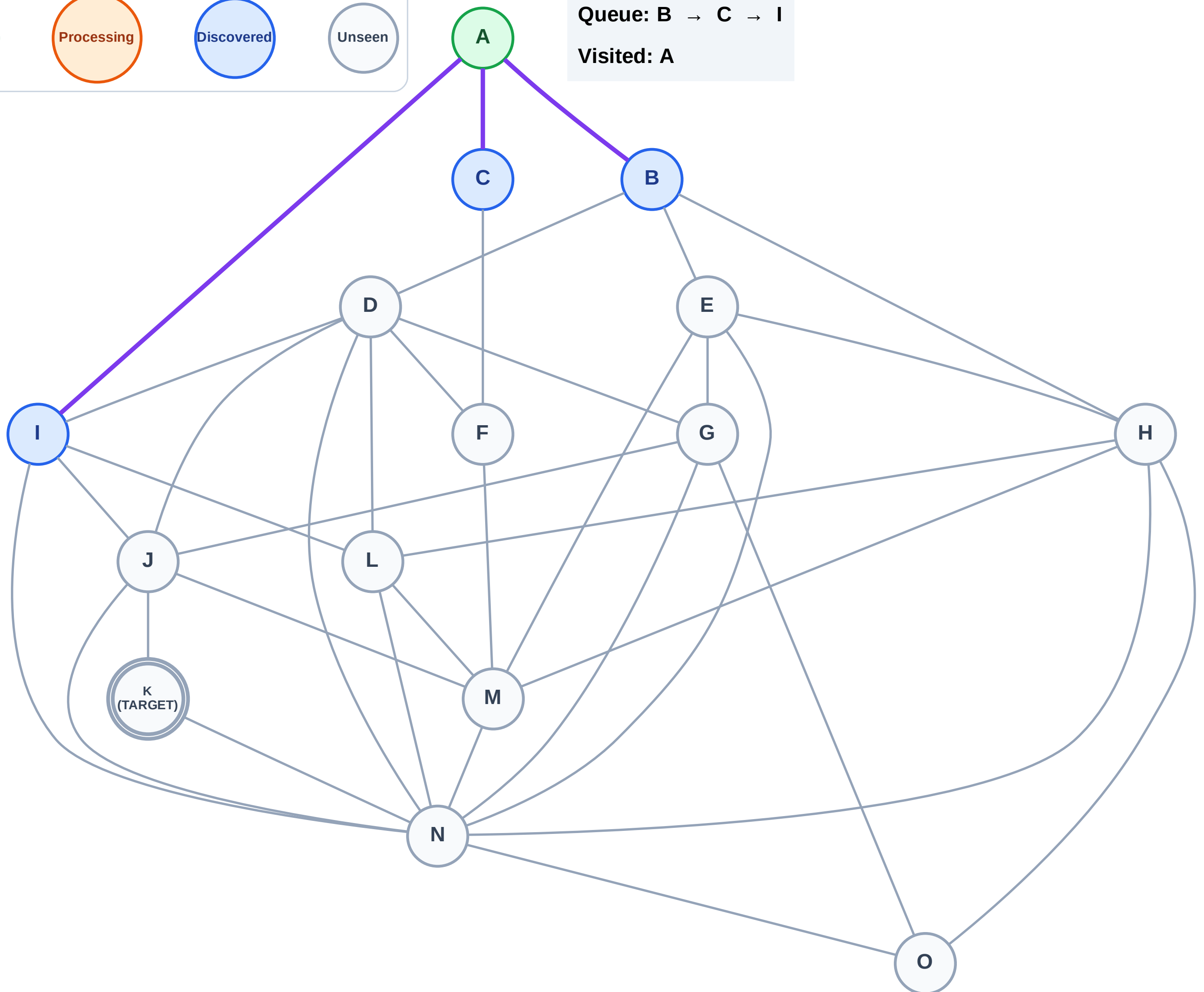
BFS Traversal

Step 3: Discover B, C, I from A



Queue: B → C → I

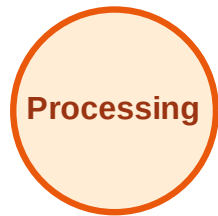
Visited: A



BFS Traversal

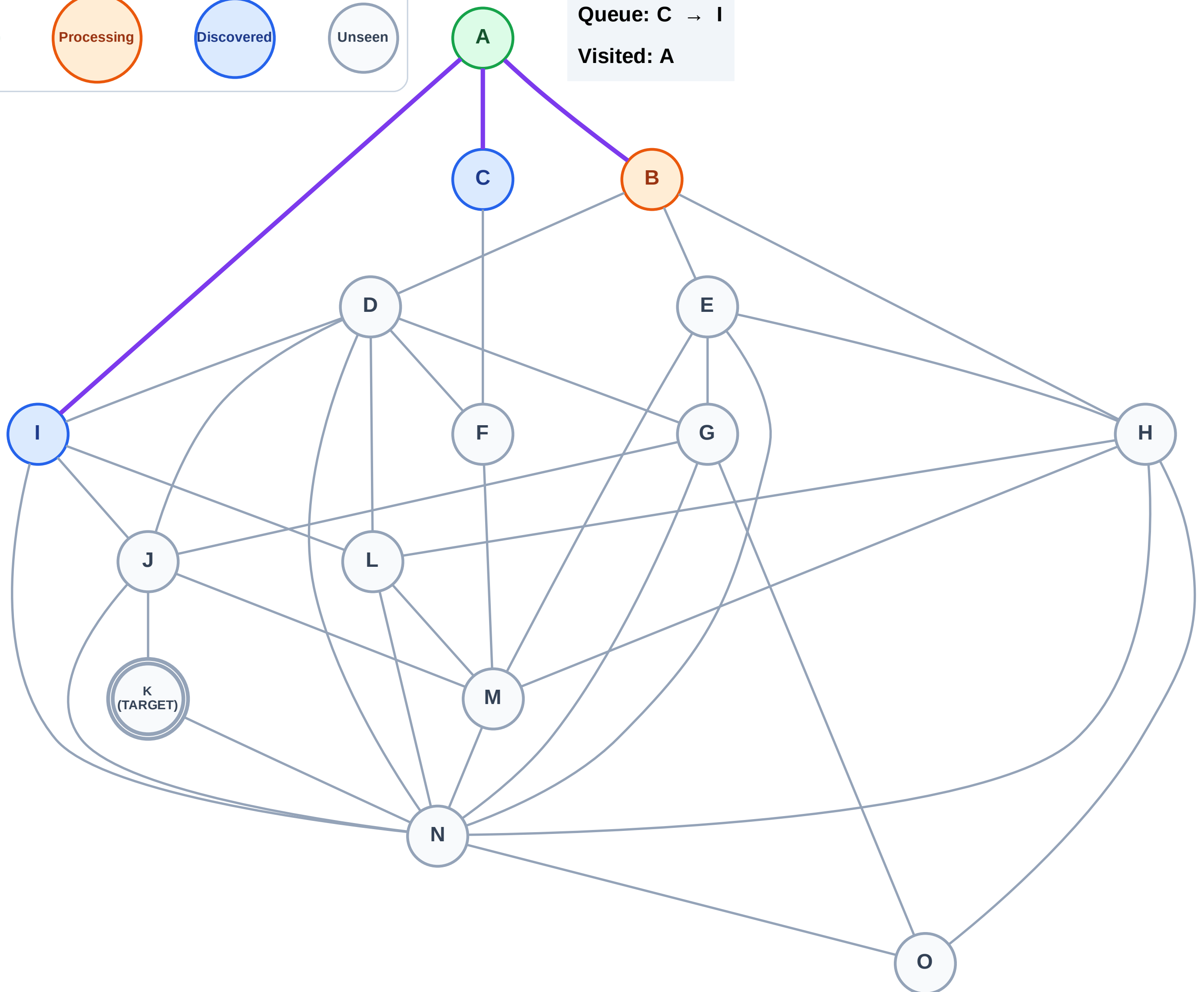
Step 4: Process node B

Legend



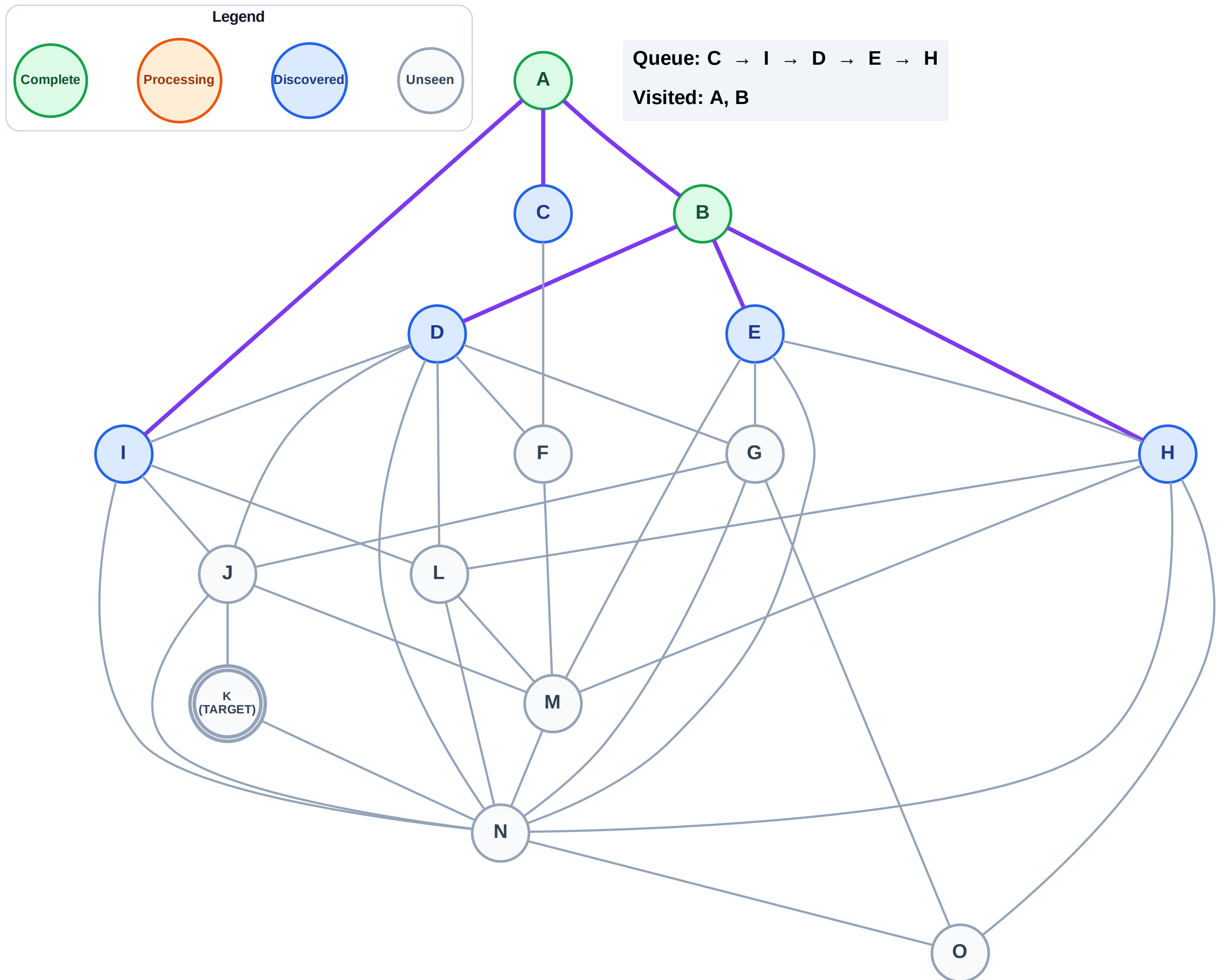
Queue: C → I

Visited: A



Step 5: Discover D, E, H from B

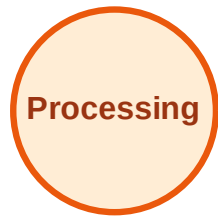
Step 5: Discover D, E, H from B



BFS Traversal

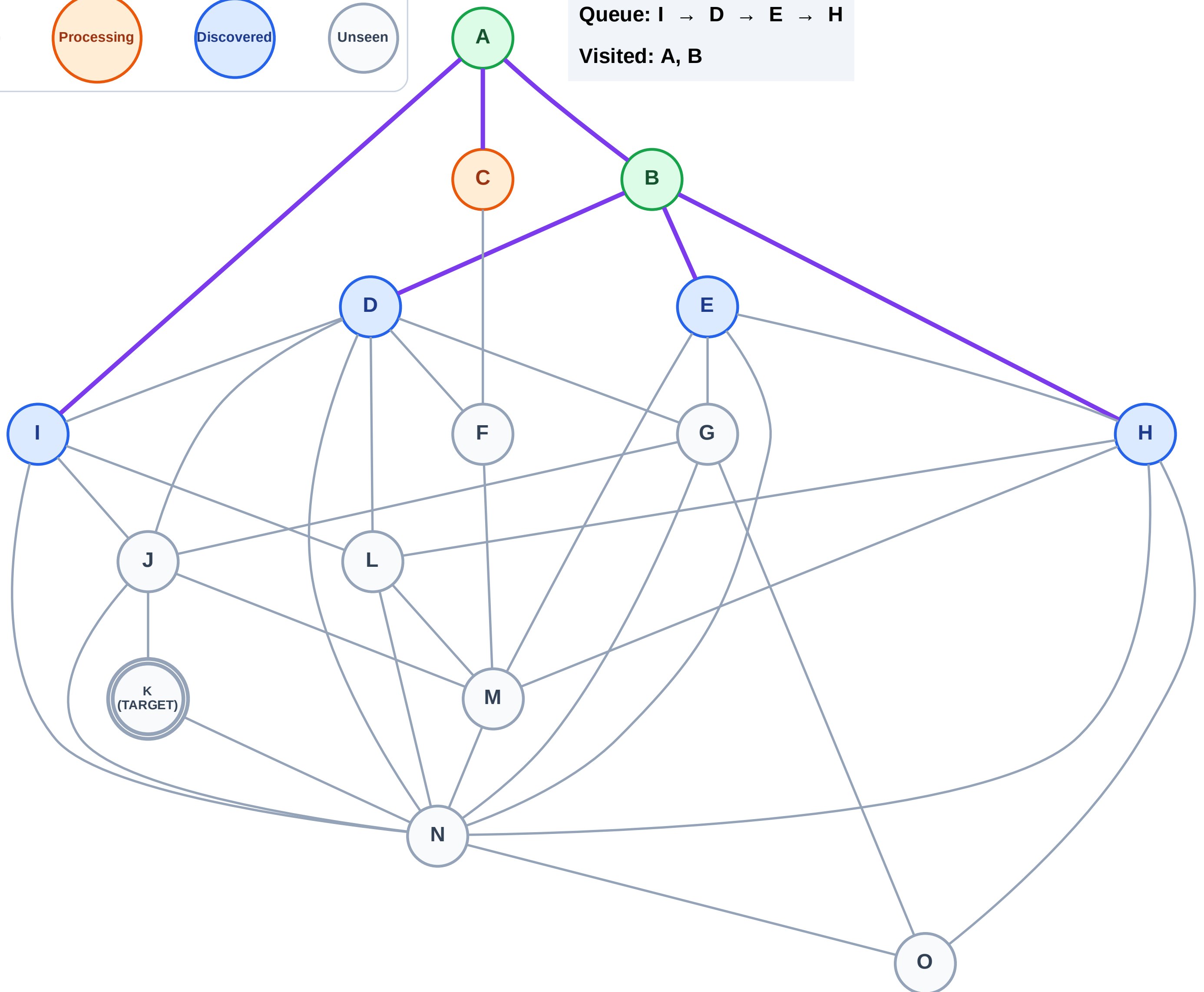
Step 6: Process node C

Legend



Queue: I → D → E → H

Visited: A, B



BFS Traversal

Step 7: Discover F from C

Legend

Complete

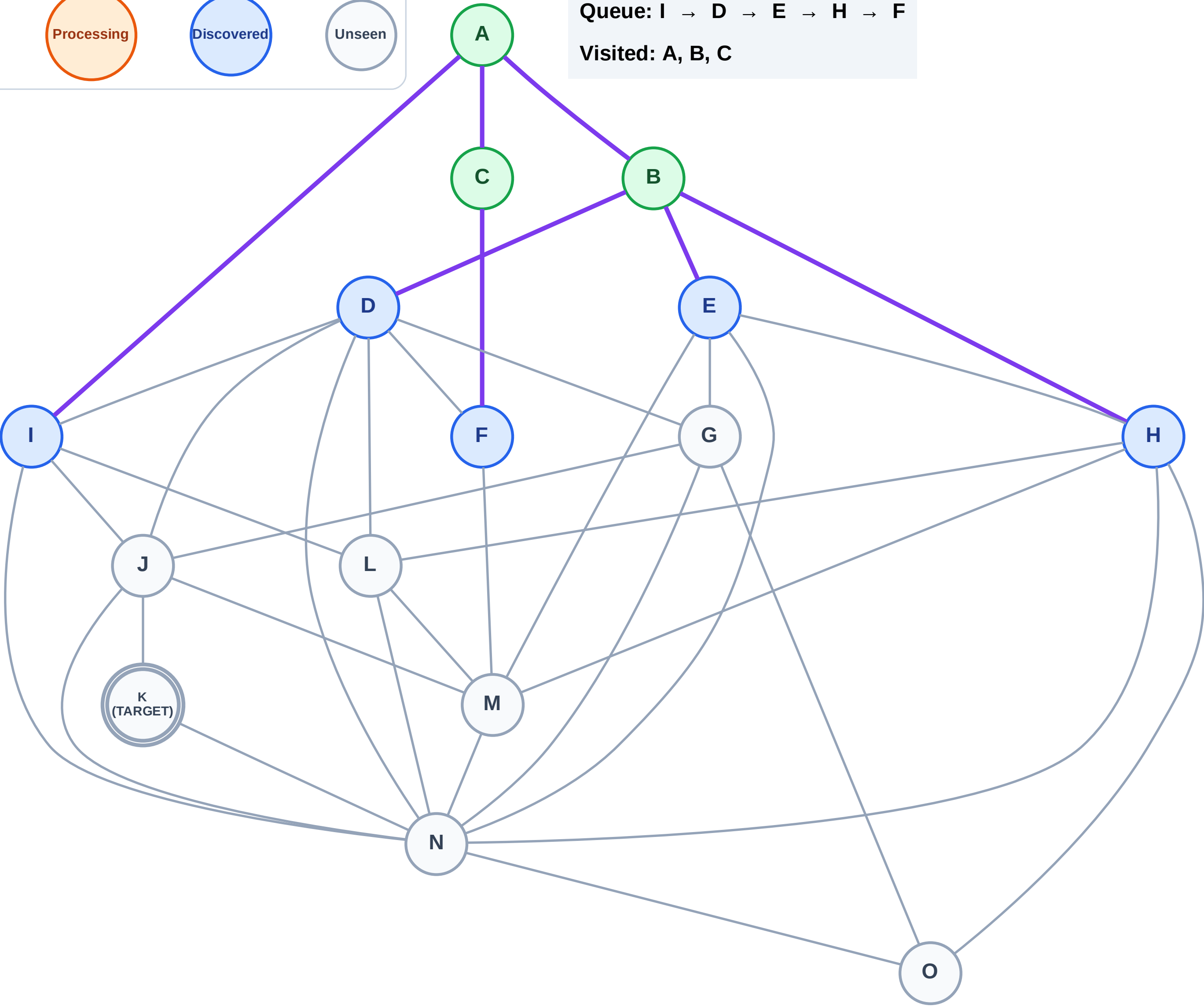
Processing

Discovered

Unseen

Queue: I → D → E → H → F

Visited: A, B, C



Step 8: Process node I

Step 8: Process node I

Complete



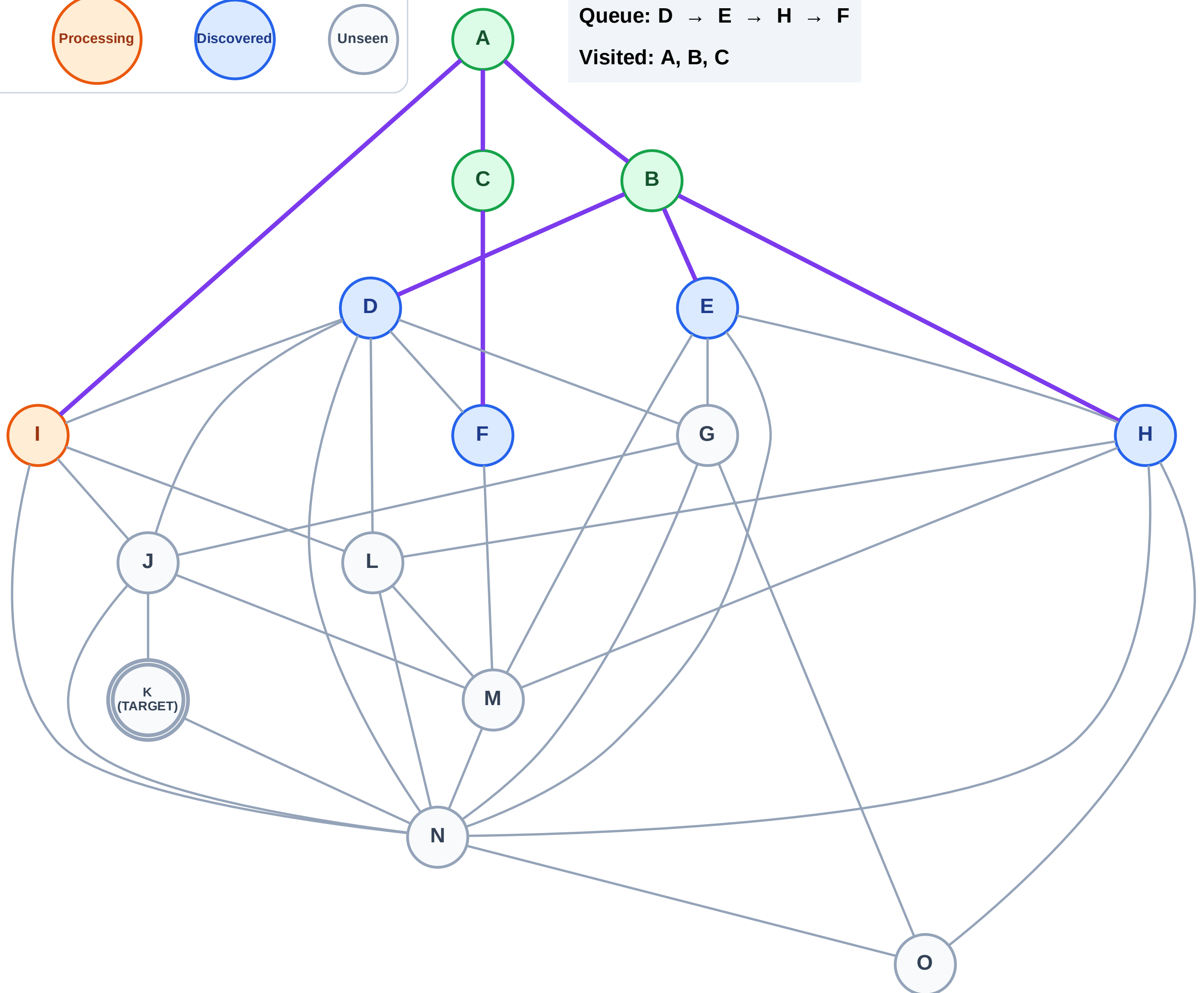
Processing



Discovered

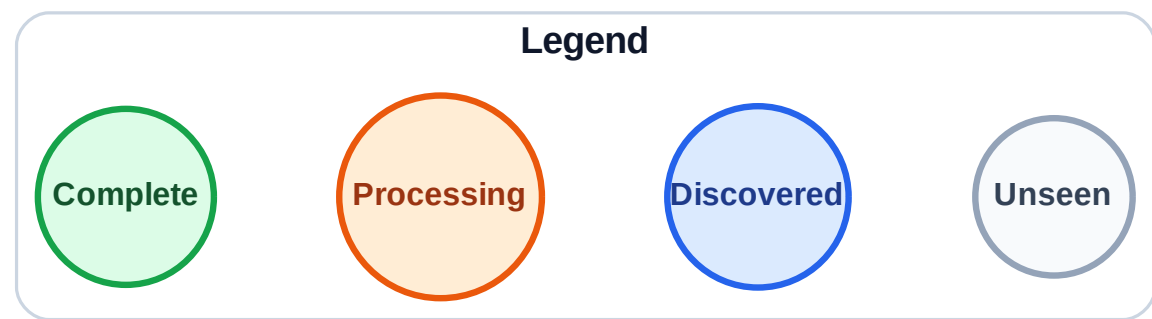
Unseen

Visited: A, B, C



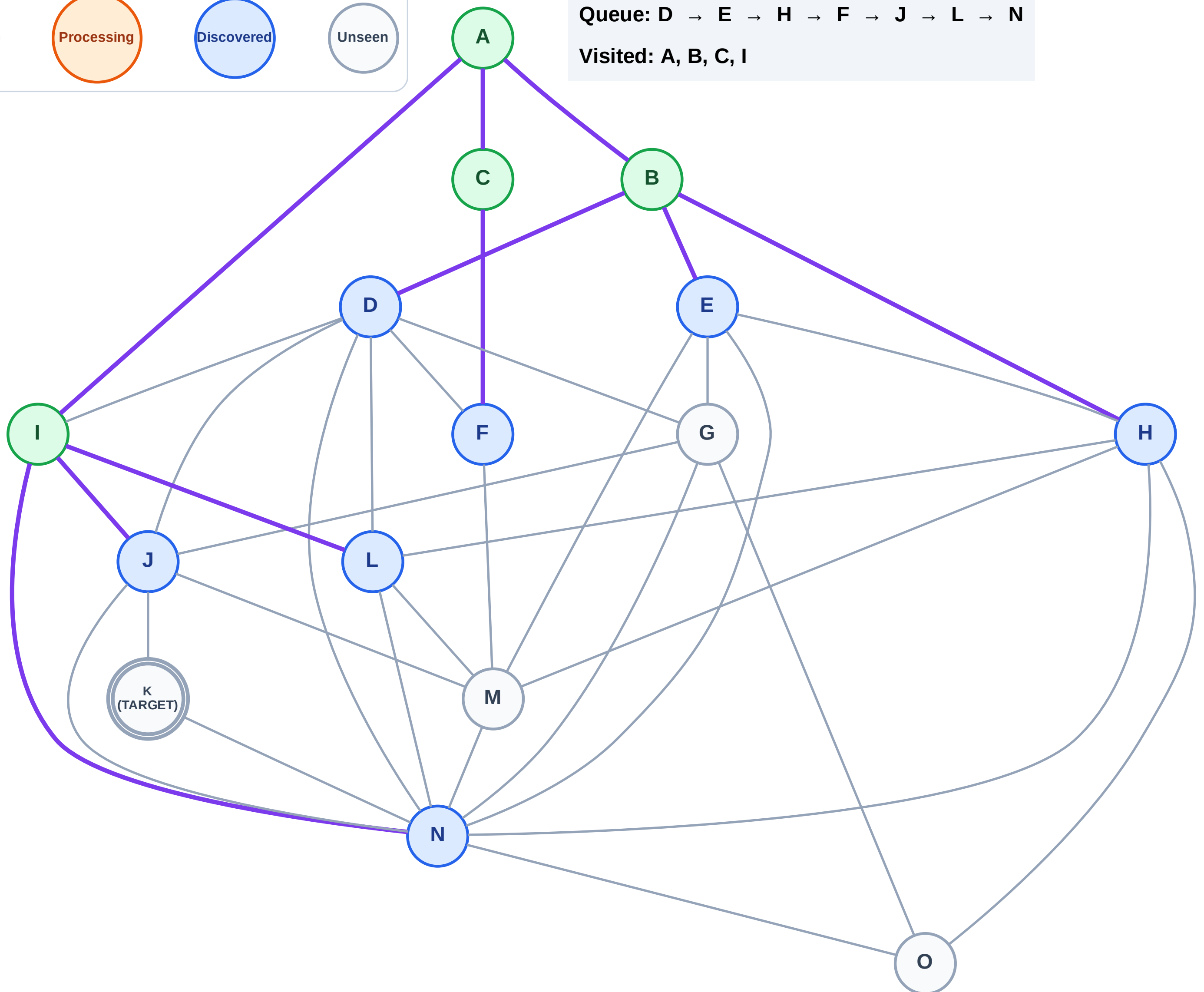
BFS Traversal

Step 9: Discover J, L, N from I



Queue: D → E → H → F → J → L → N

Visited: A, B, C, I



BFS Traversal

Step 10: Process node D

Legend

Complete

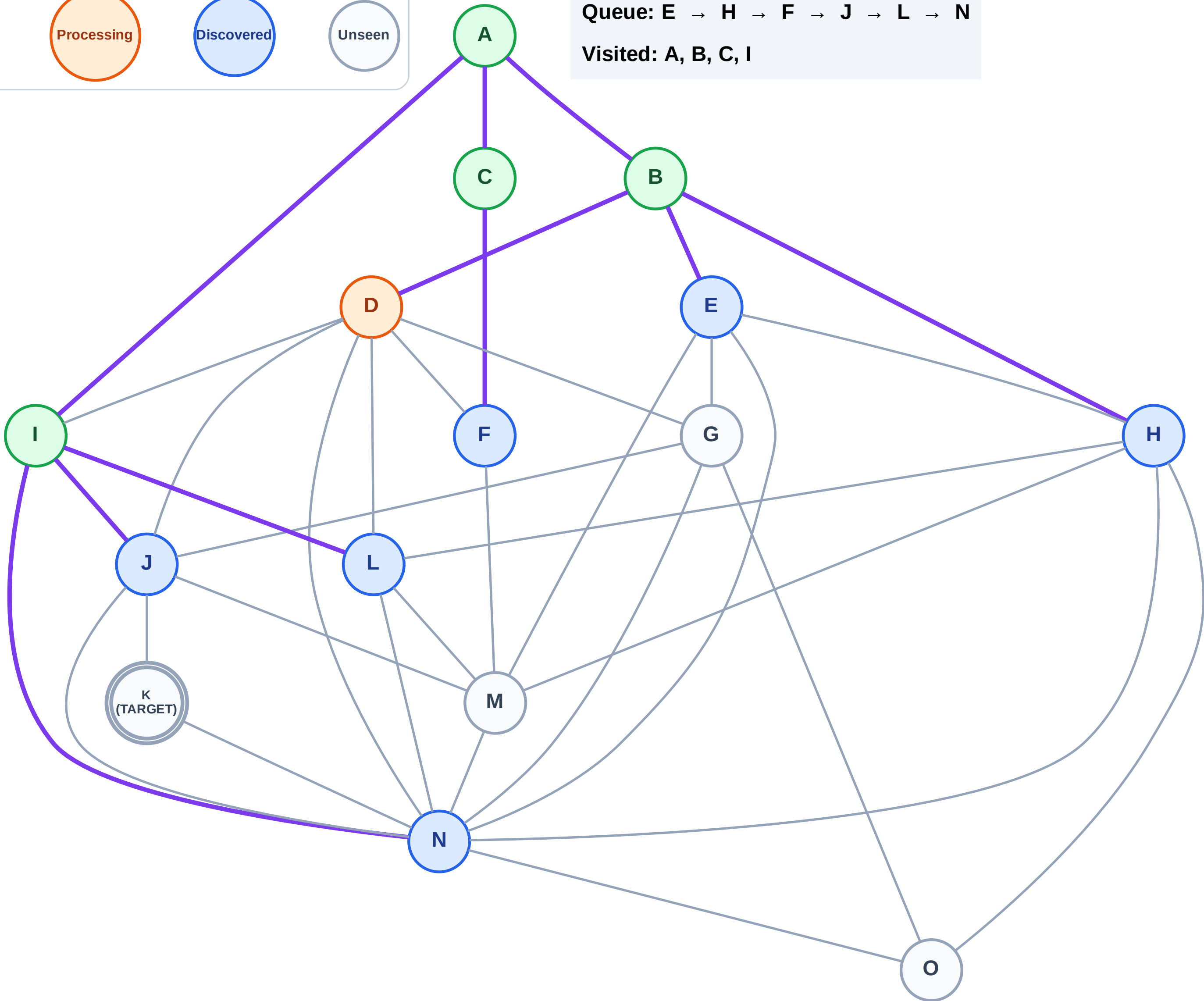
Processing

Discovered

Unseen

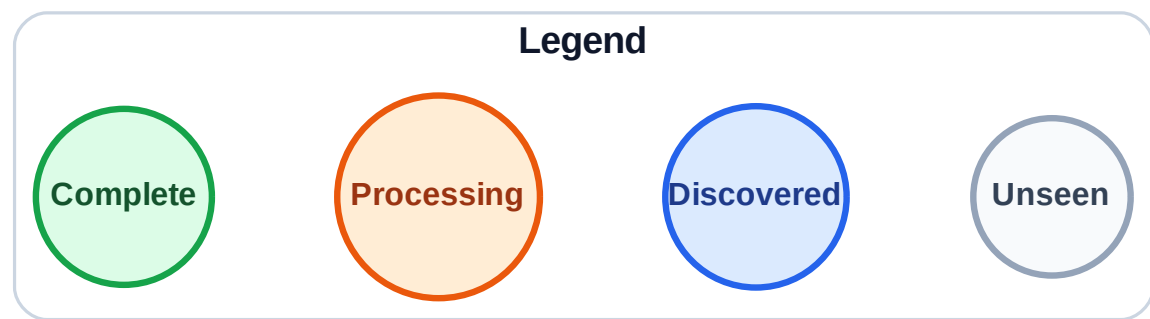
Queue: E → H → F → J → L → N

Visited: A, B, C, I



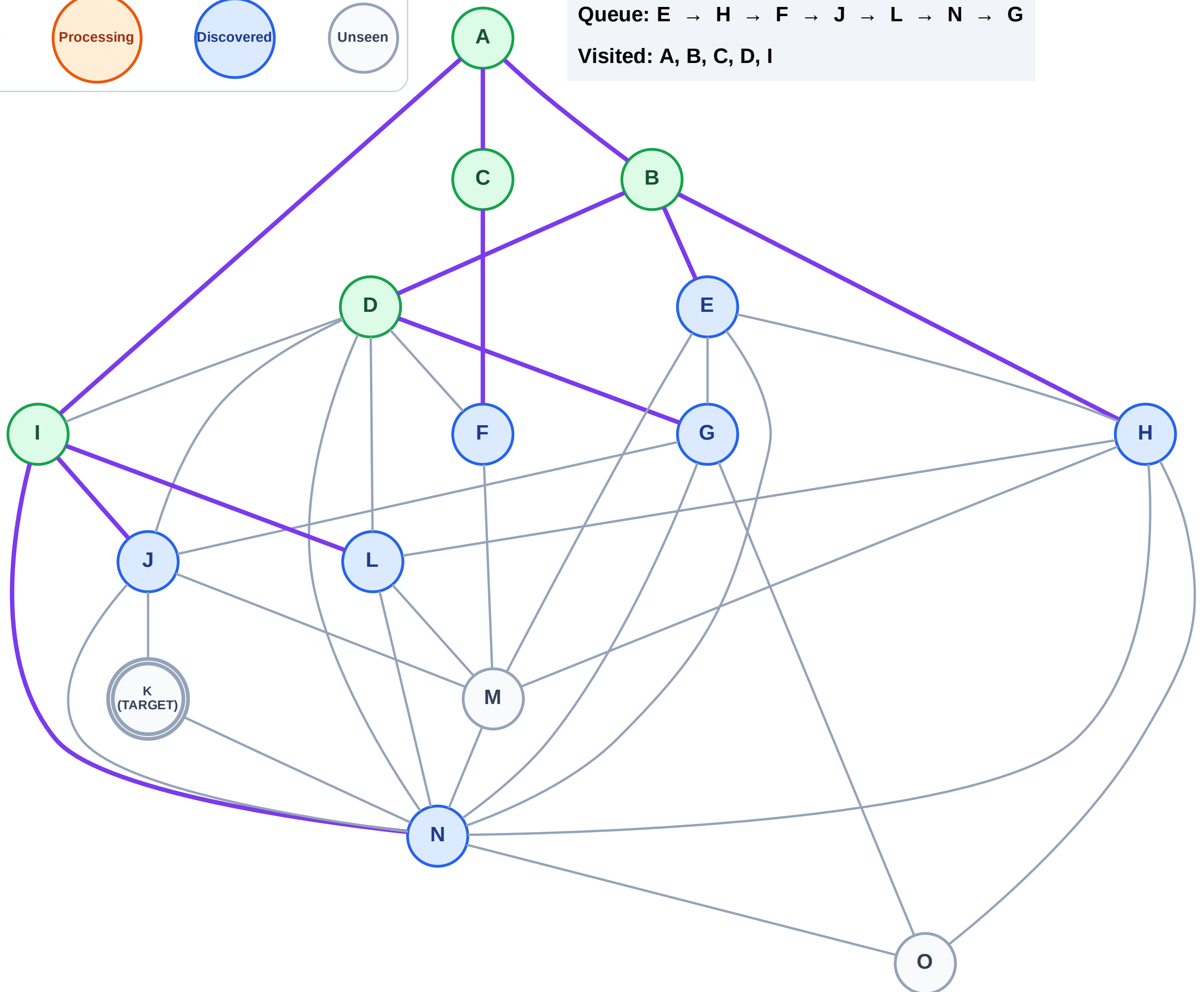
BFS Traversal

Step 11: Discover G from D



Queue: E → H → F → J → L → N → G

Visited: A, B, C, D, I



BFS Traversal

Step 12: Process node E

Legend

Complete

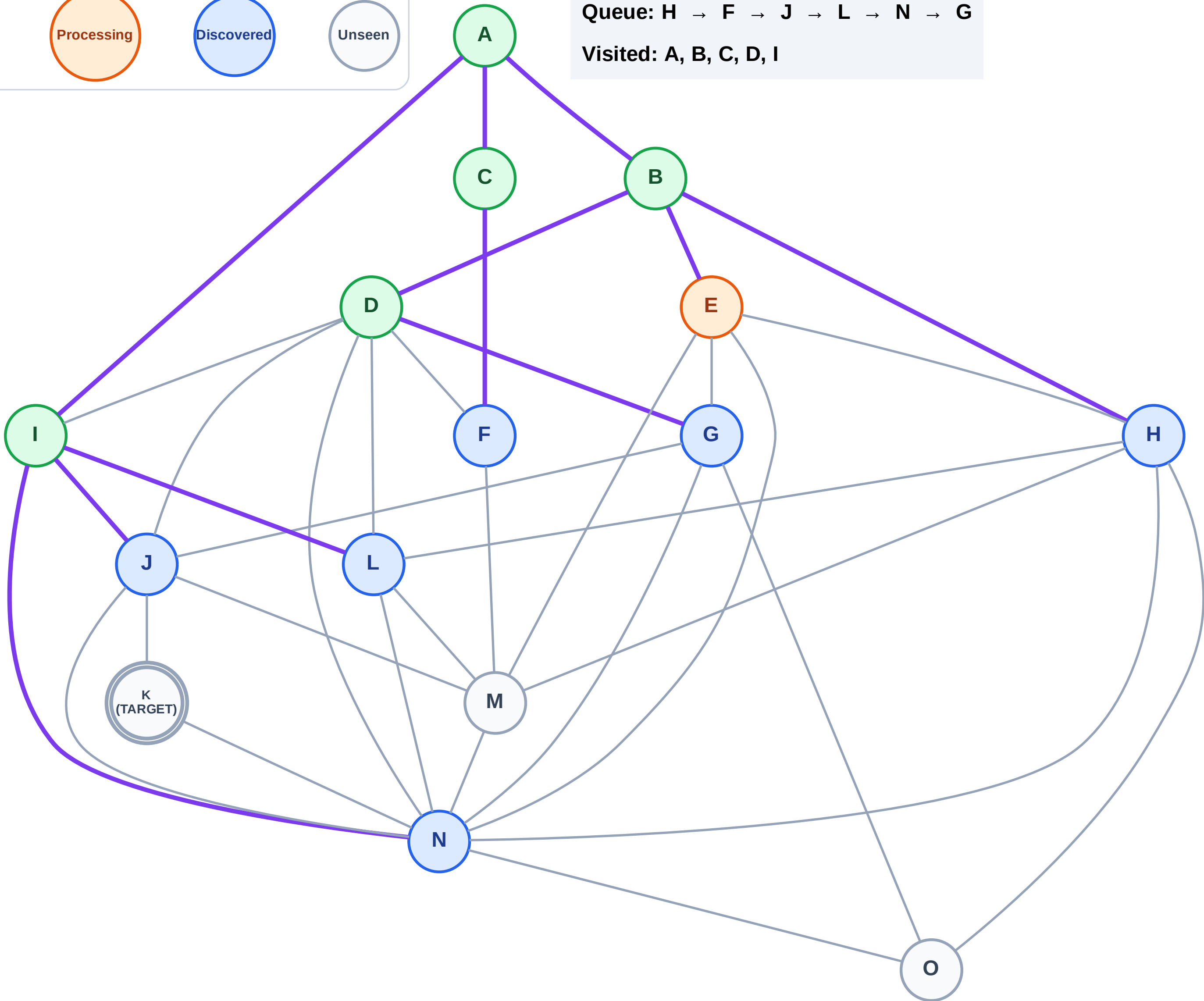
Processing

Discovered

Unseen

Queue: H → F → J → L → N → G

Visited: A, B, C, D, I



BFS Traversal

Step 13: Discover M from E

Legend



Complete



Processing



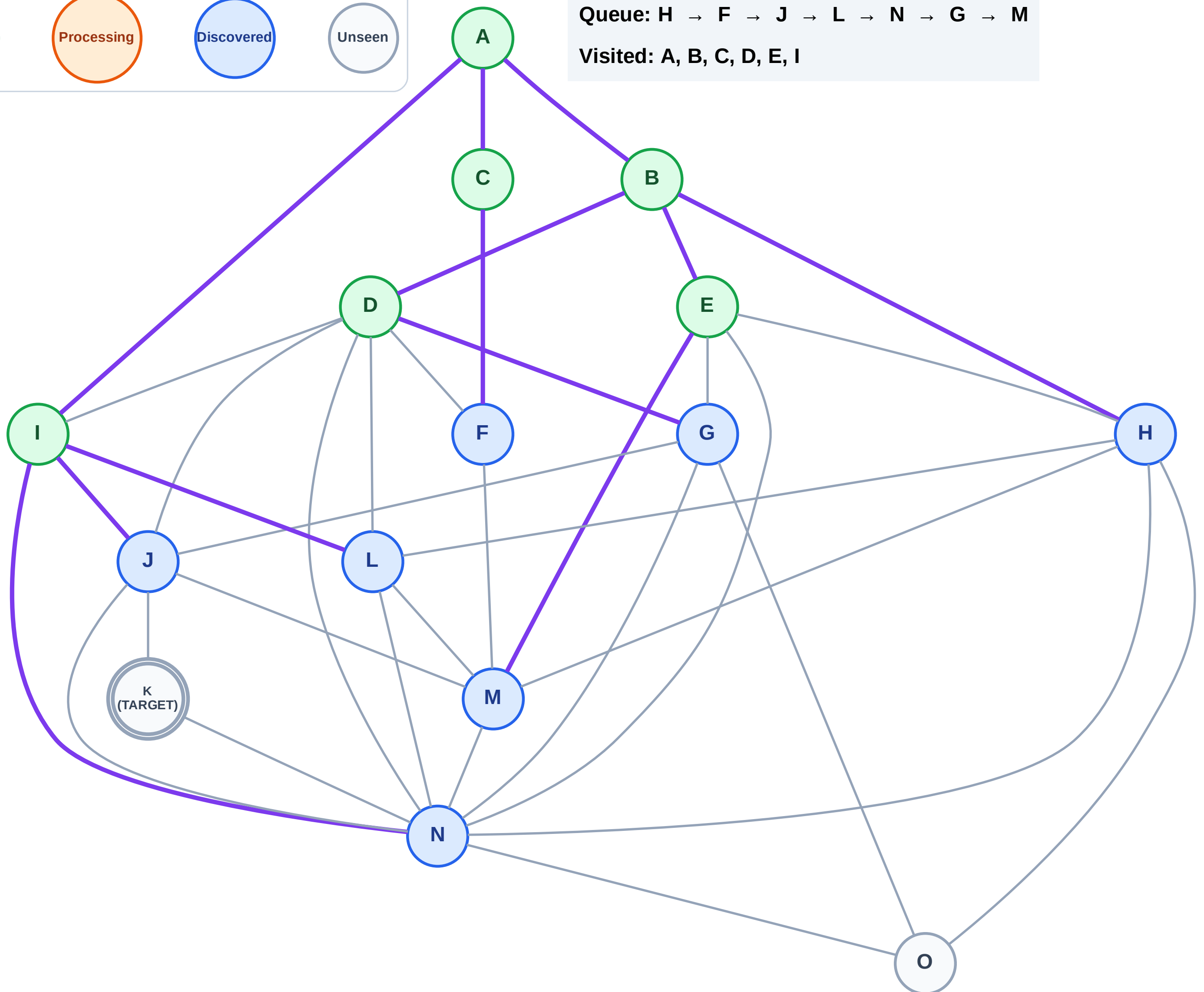
Discovered



Unseen

Queue: H → F → J → L → N → G → M

Visited: A, B, C, D, E, I



BFS Traversal

Step 14: Process node H

Legend

Complete

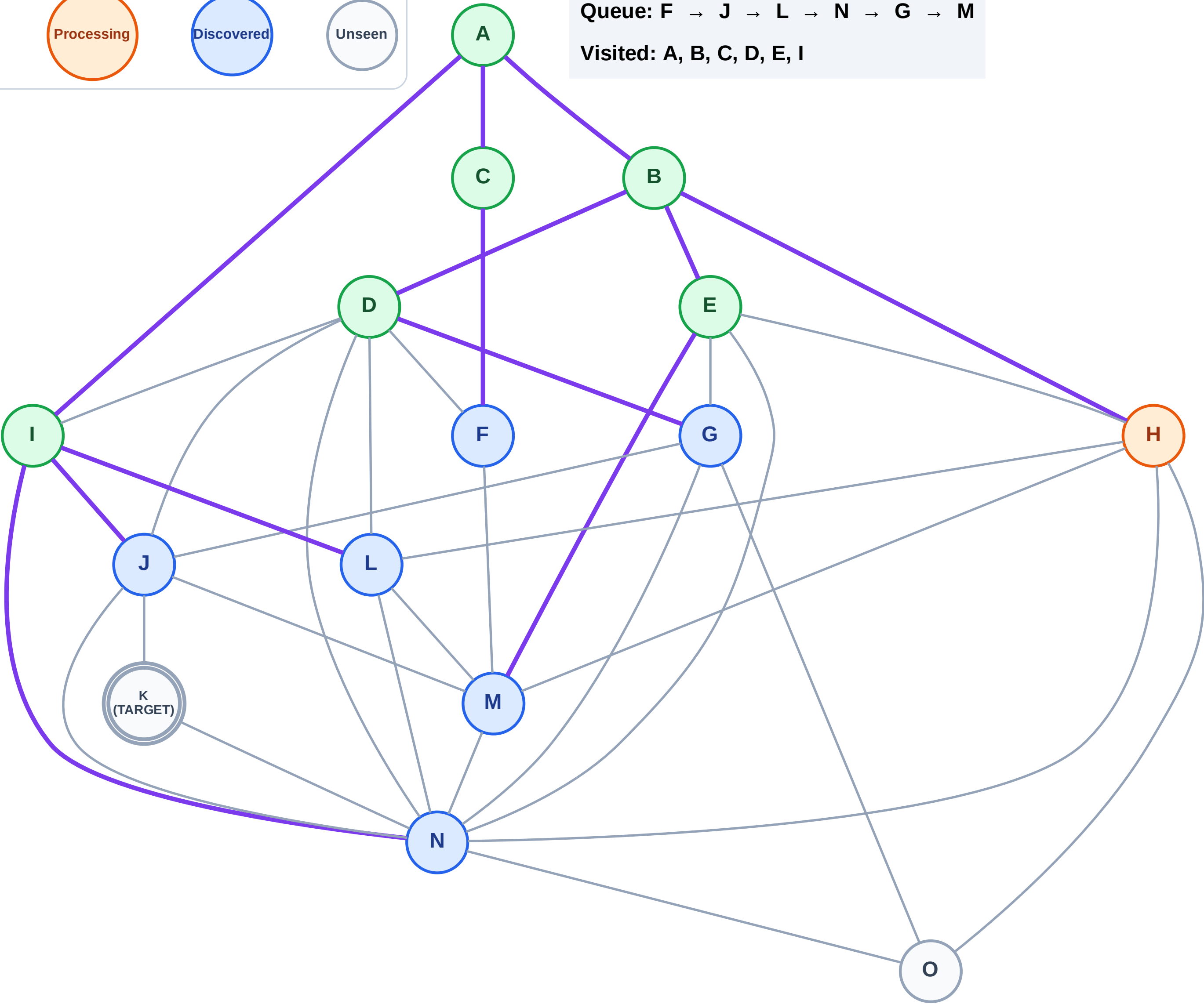
Processing

Discovered

Unseen

Queue: F → J → L → N → G → M

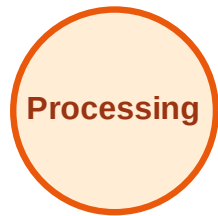
Visited: A, B, C, D, E, I



BFS Traversal

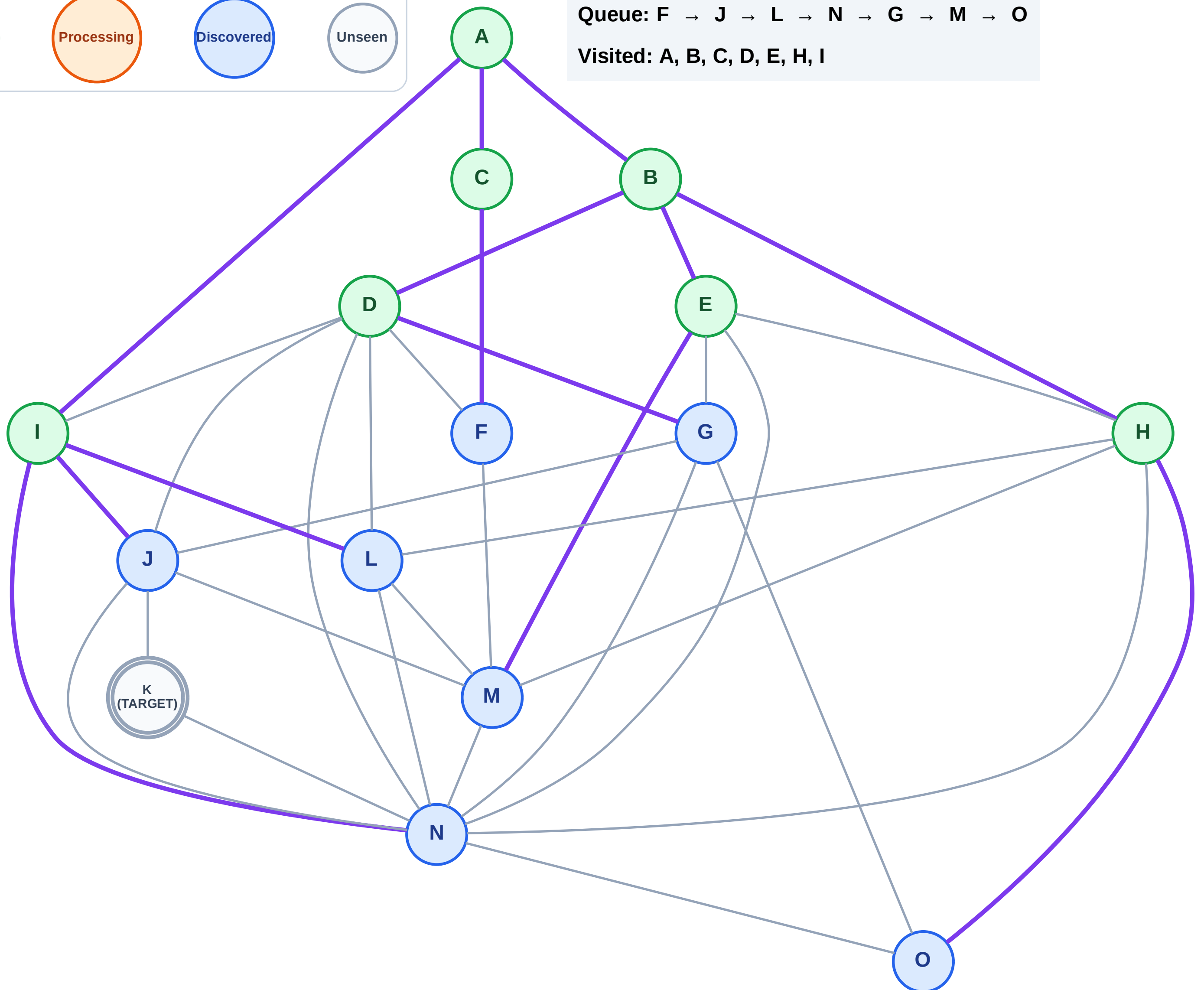
Step 15: Discover O from H

Legend



Queue: F → J → L → N → G → M → O

Visited: A, B, C, D, E, H, I



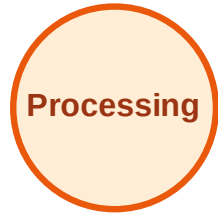
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



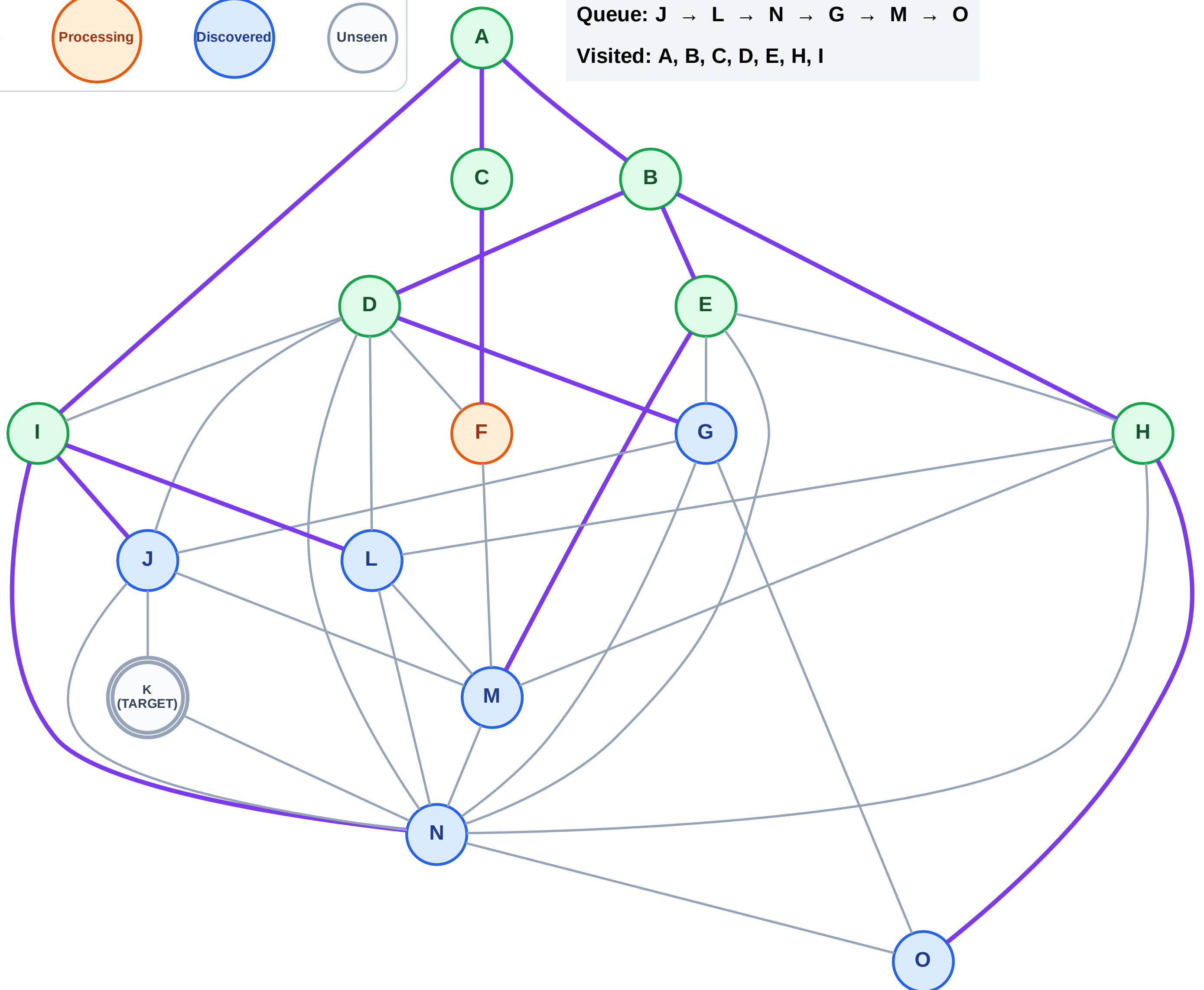
Discovered



Unseen

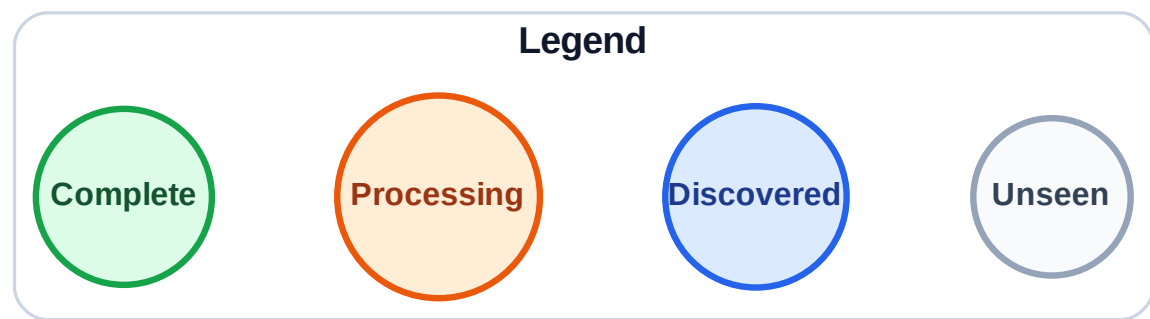
Queue: J → L → N → G → M → O

Visited: A, B, C, D, E, H, I



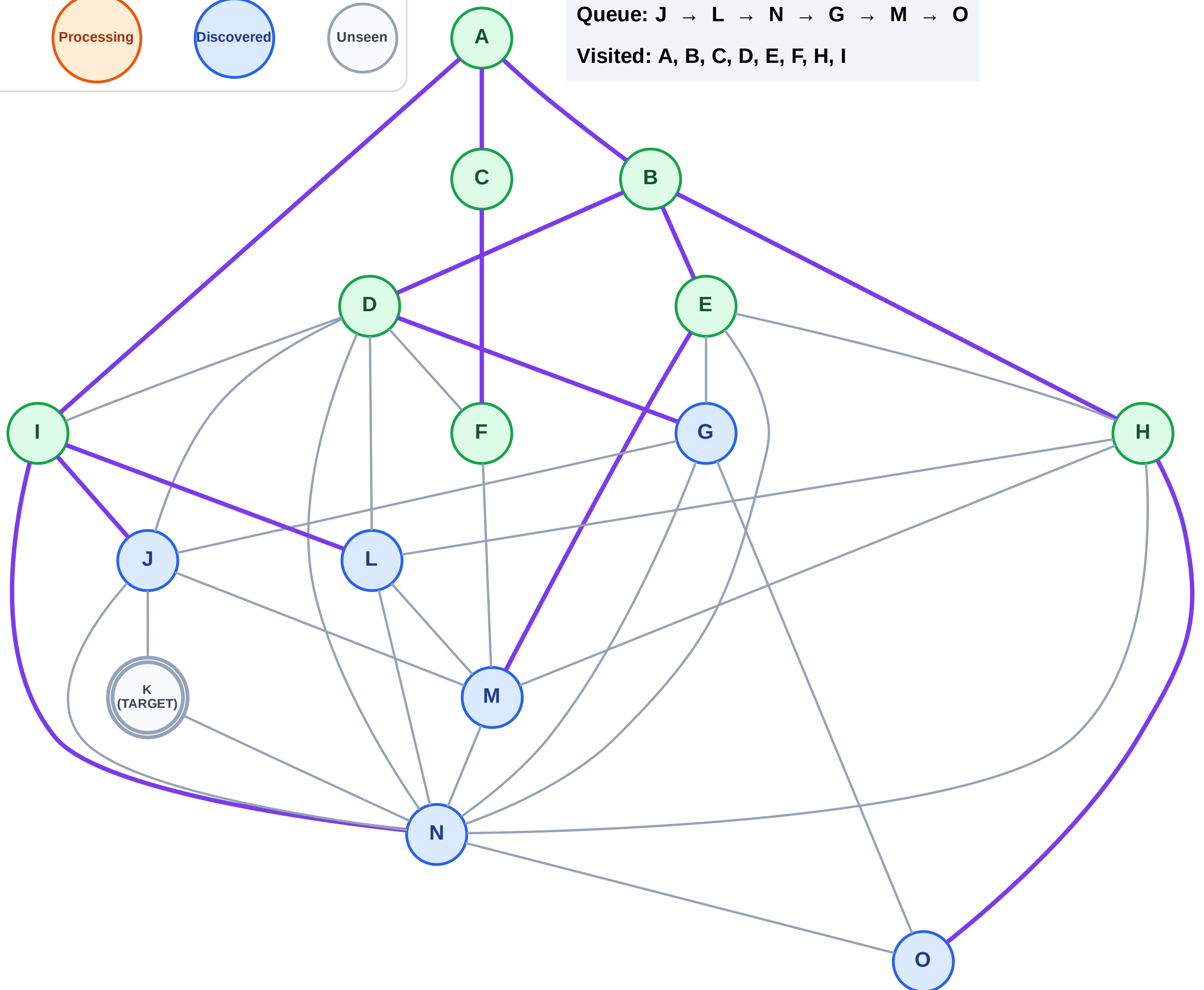
BFS Traversal

Step 17: No new neighbors from F



Queue: J → L → N → G → M → O

Visited: A, B, C, D, E, F, H, I



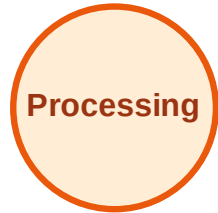
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



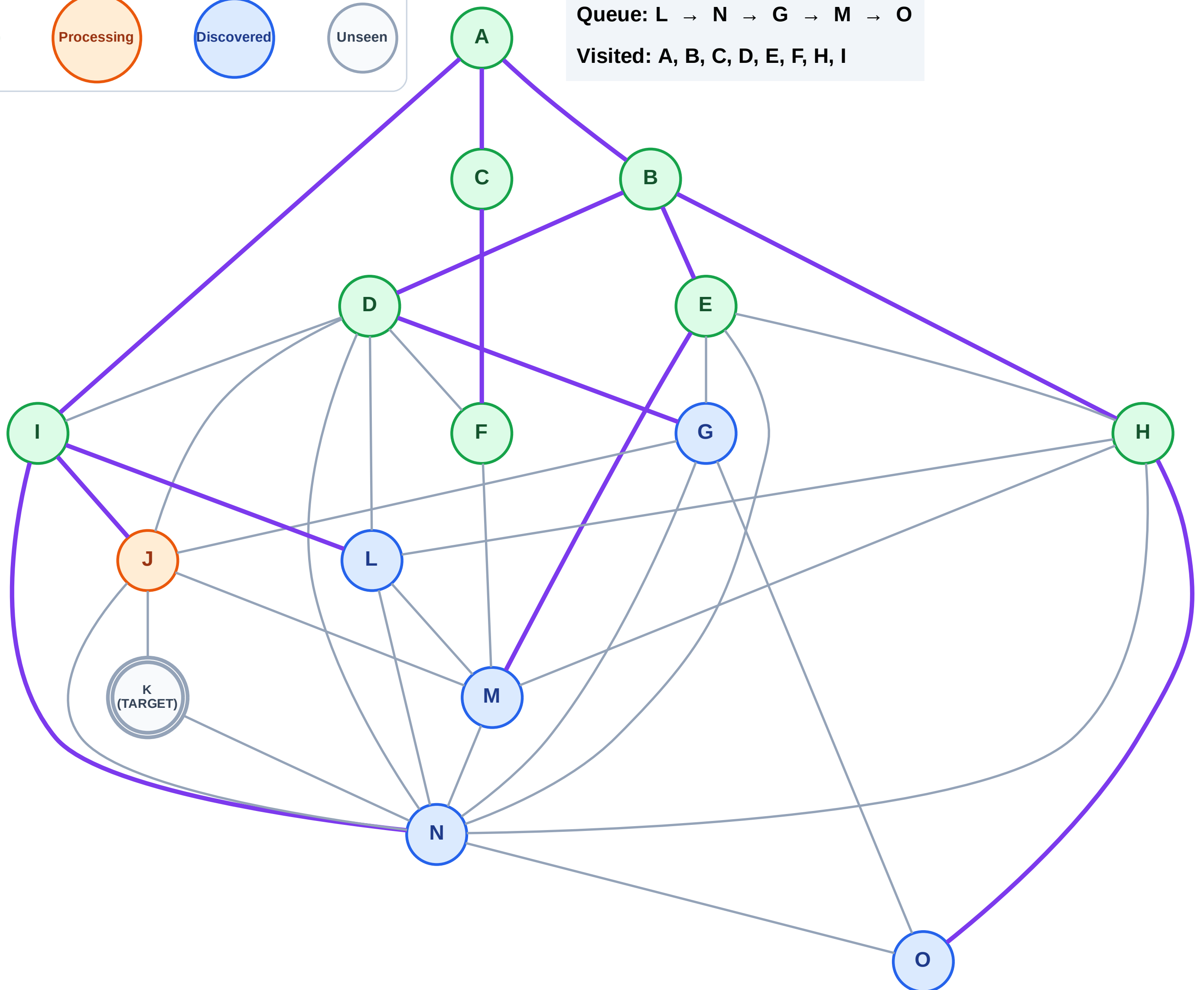
Discovered



Unseen

Queue: L → N → G → M → O

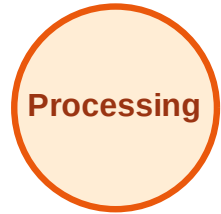
Visited: A, B, C, D, E, F, H, I



BFS Traversal

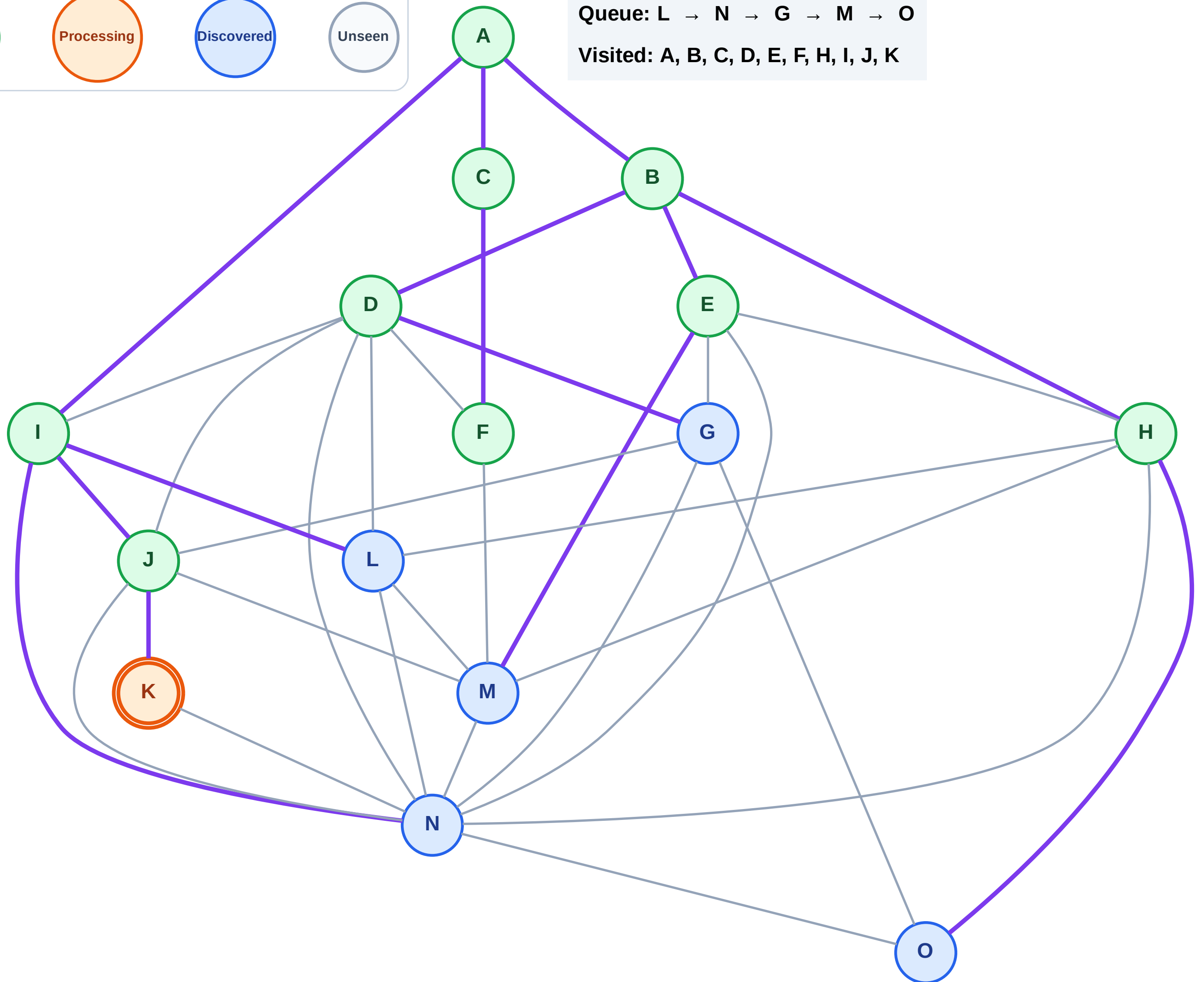
Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → I → J → K

Legend



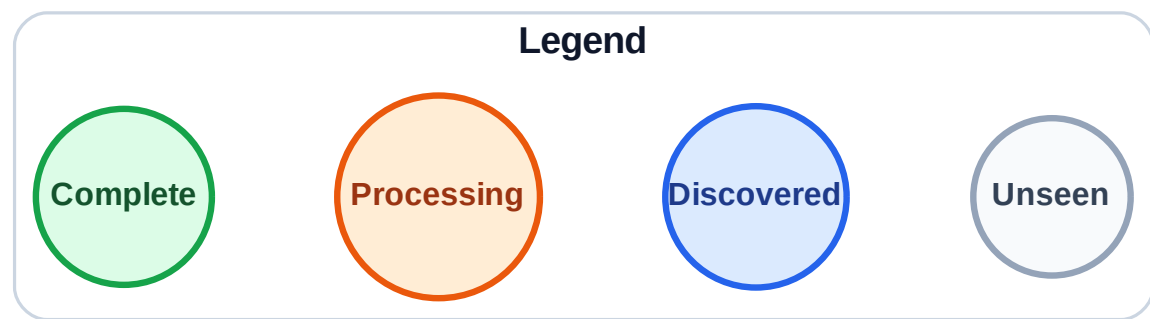
Queue: L → N → G → M → O

Visited: A, B, C, D, E, F, H, I, J, K



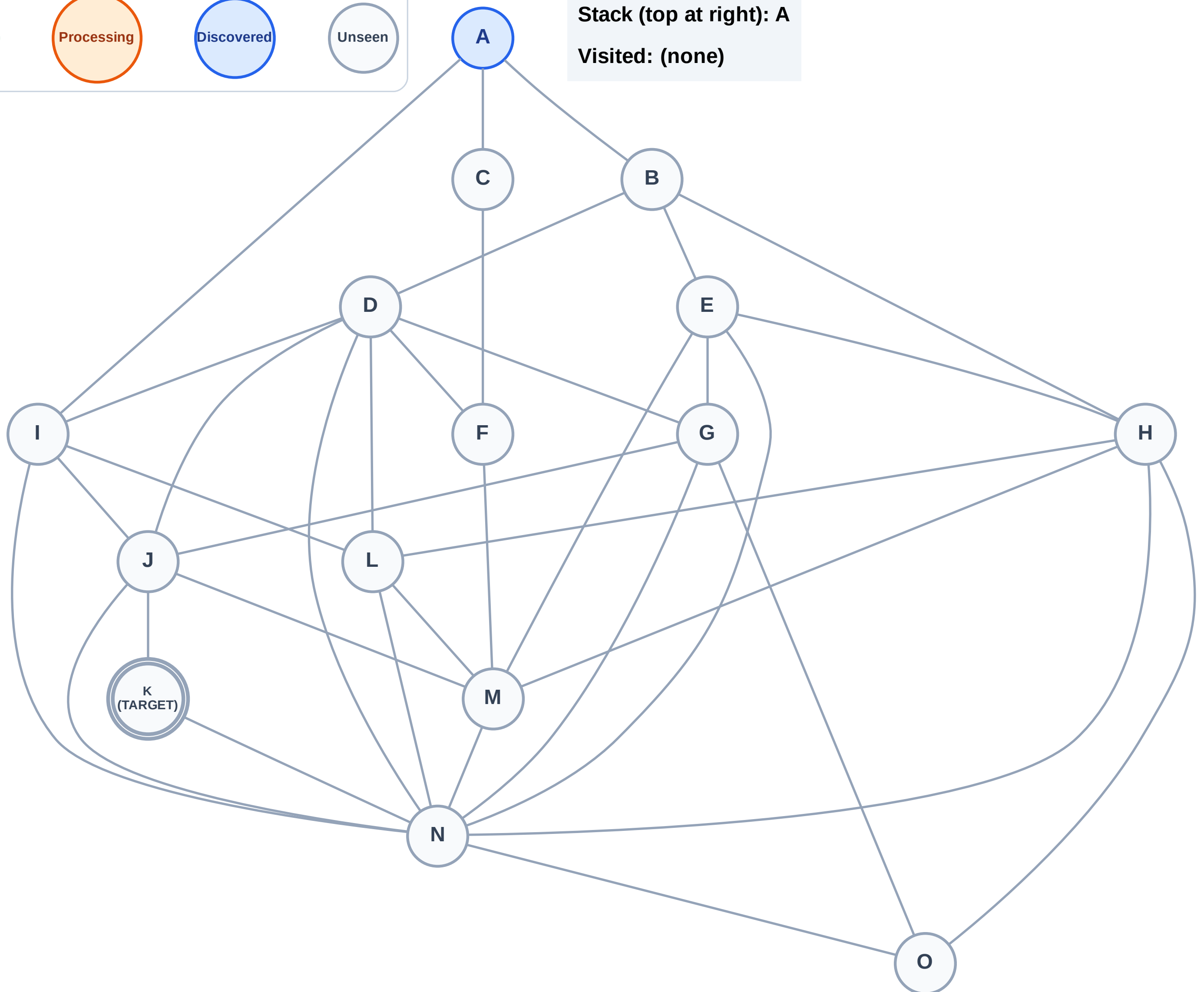
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

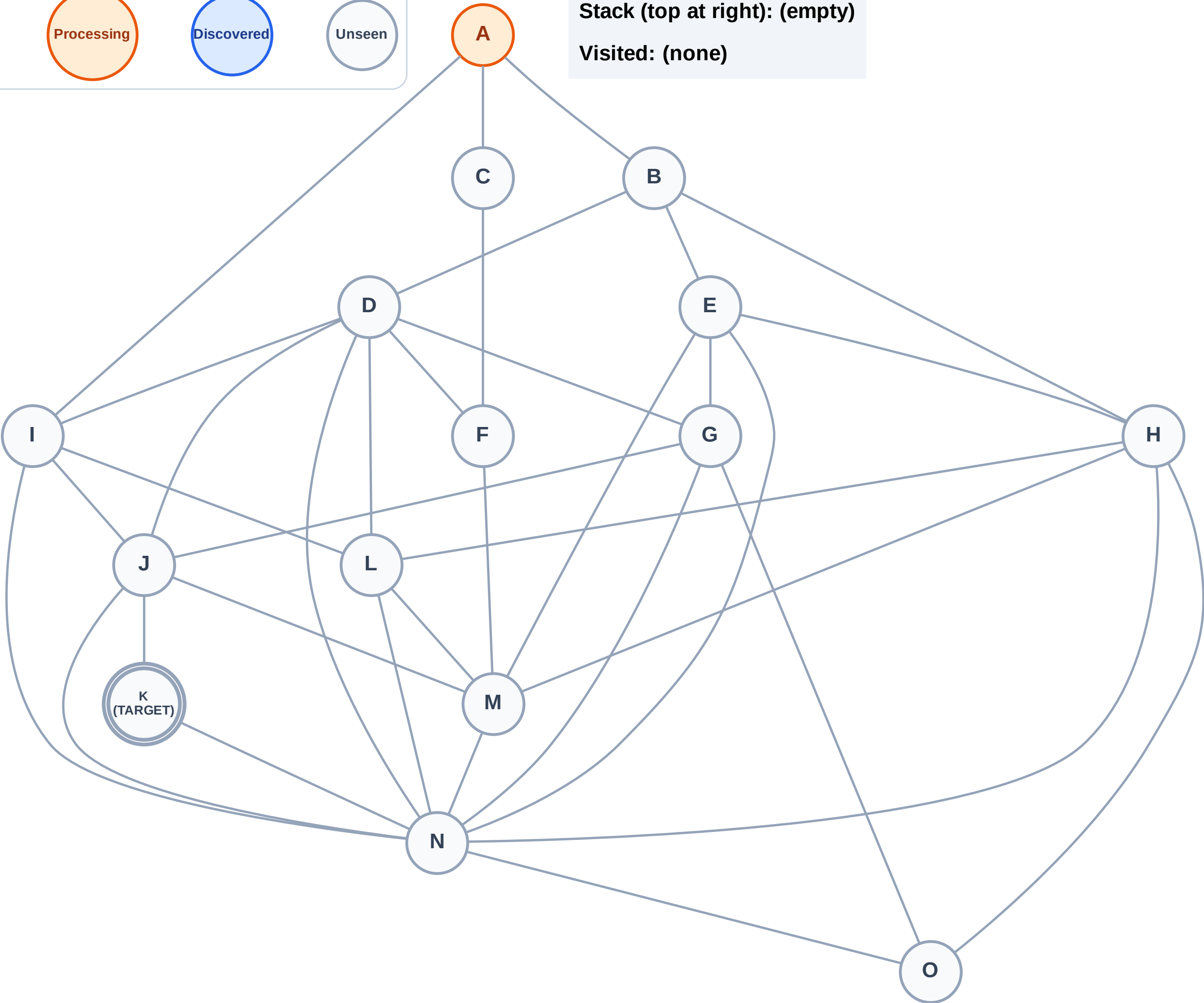
Complete

Processing

Discovered

Unseen

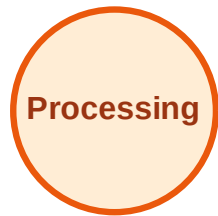
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

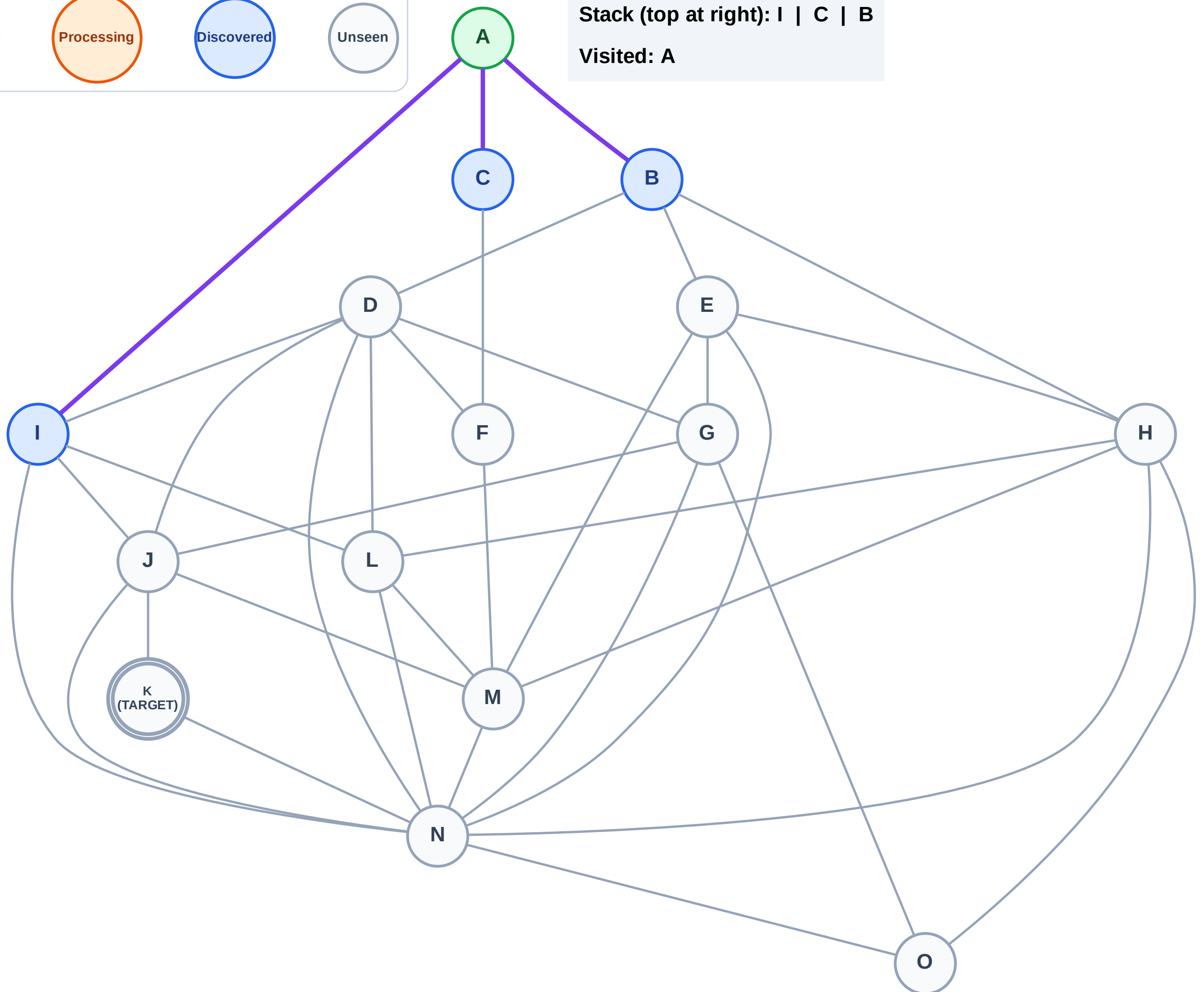
Step 3: Discover I, C, B from A

Legend



Stack (top at right): I | C | B

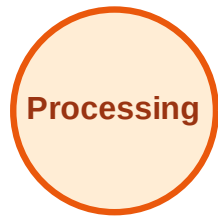
Visited: A



DFS Traversal

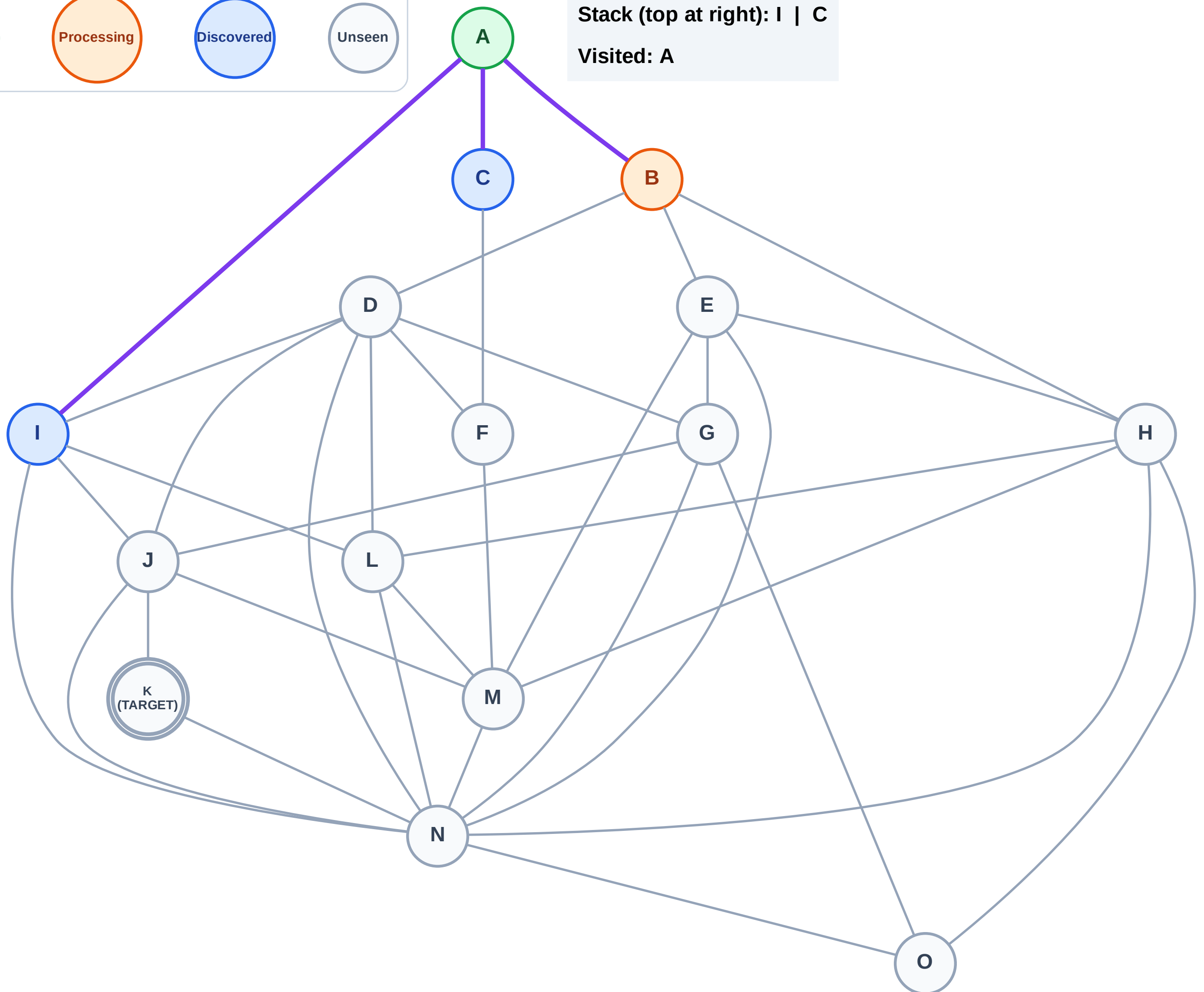
Step 4: Process node B

Legend



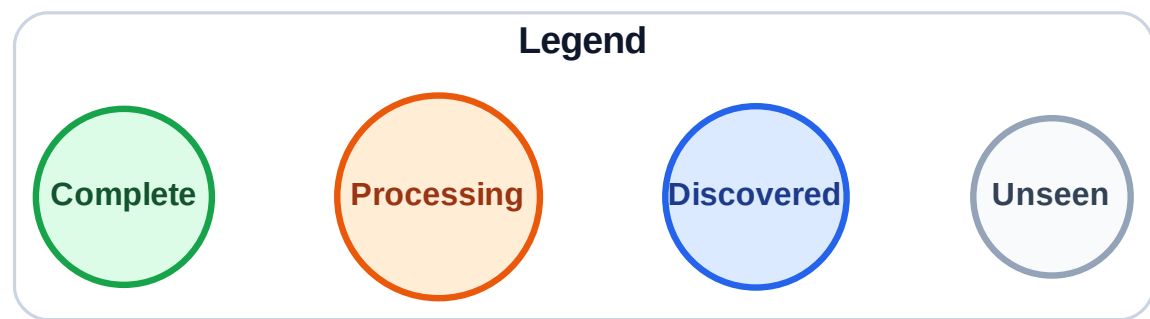
Stack (top at right): I | C

Visited: A



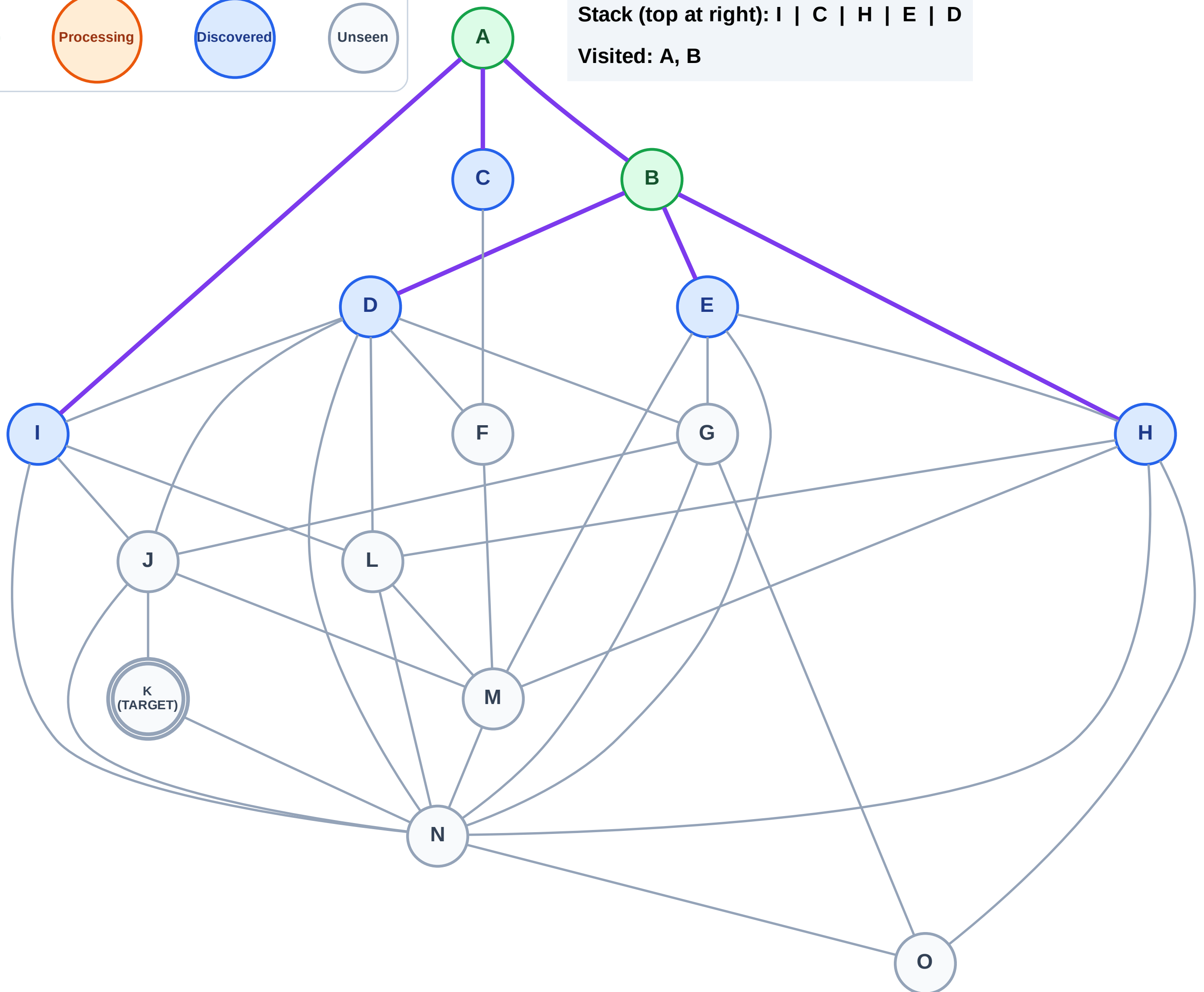
DFS Traversal

Step 5: Discover H, E, D from B



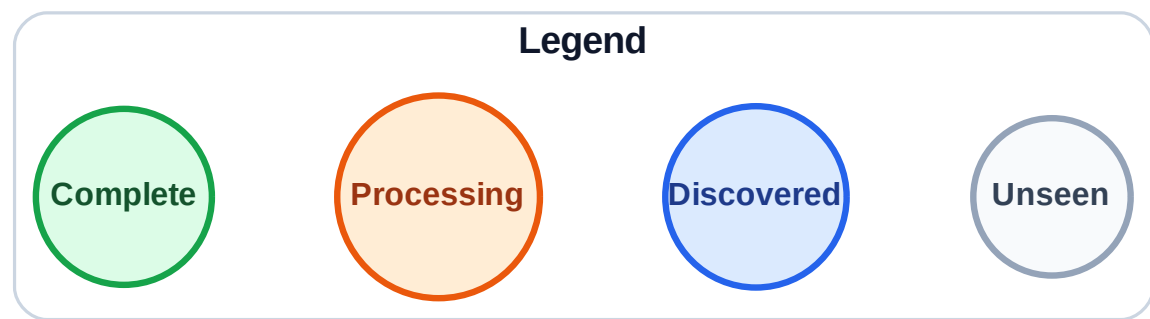
Stack (top at right): I | C | H | E | D

Visited: A, B



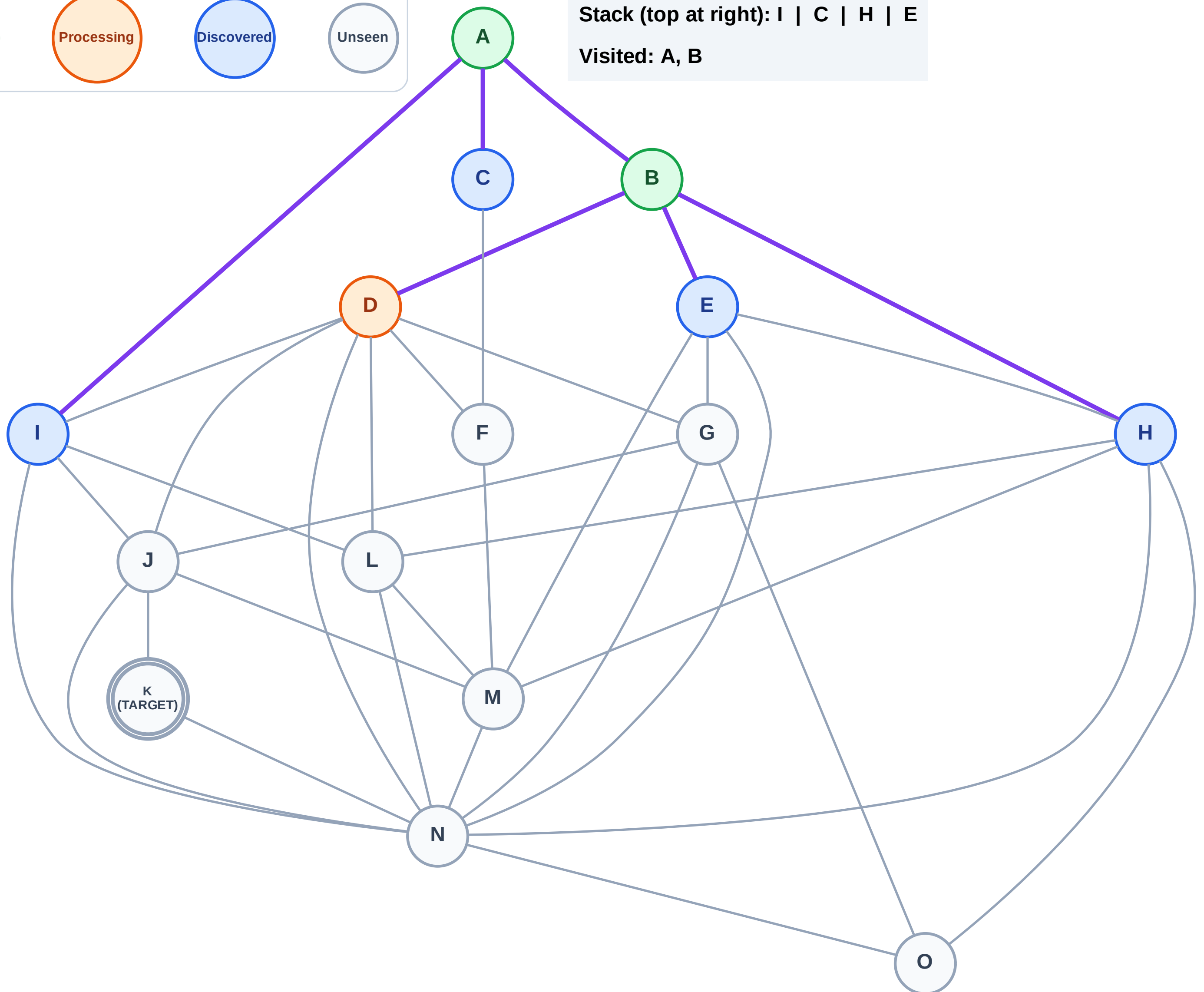
DFS Traversal

Step 6: Process node D



Stack (top at right): I | C | H | E

Visited: A, B



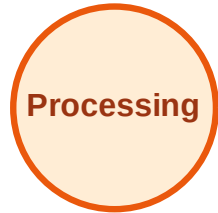
DFS Traversal

Step 7: Discover N, L, J, G, F from D

Legend



Complete



Processing



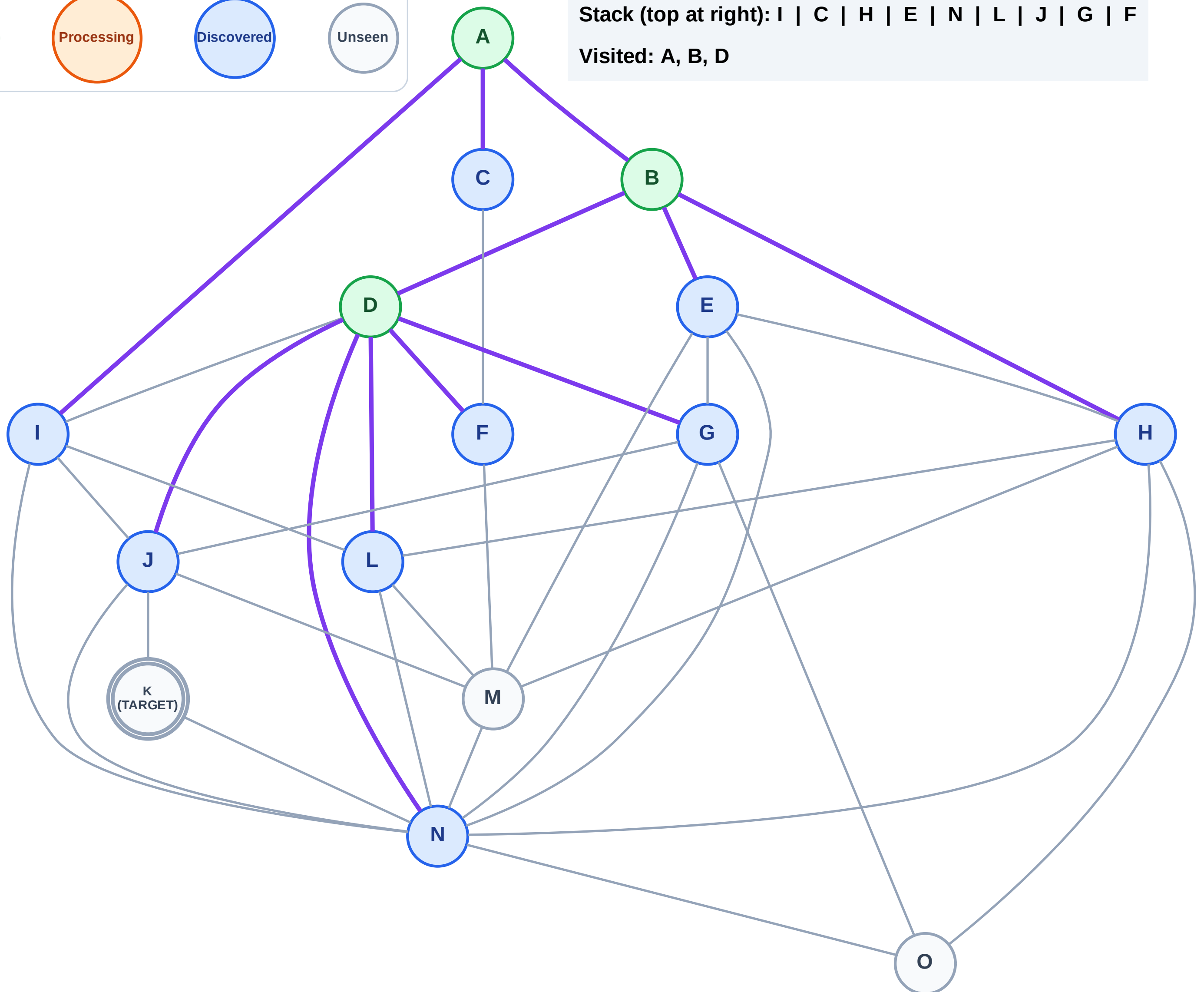
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L | J | G | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

Complete

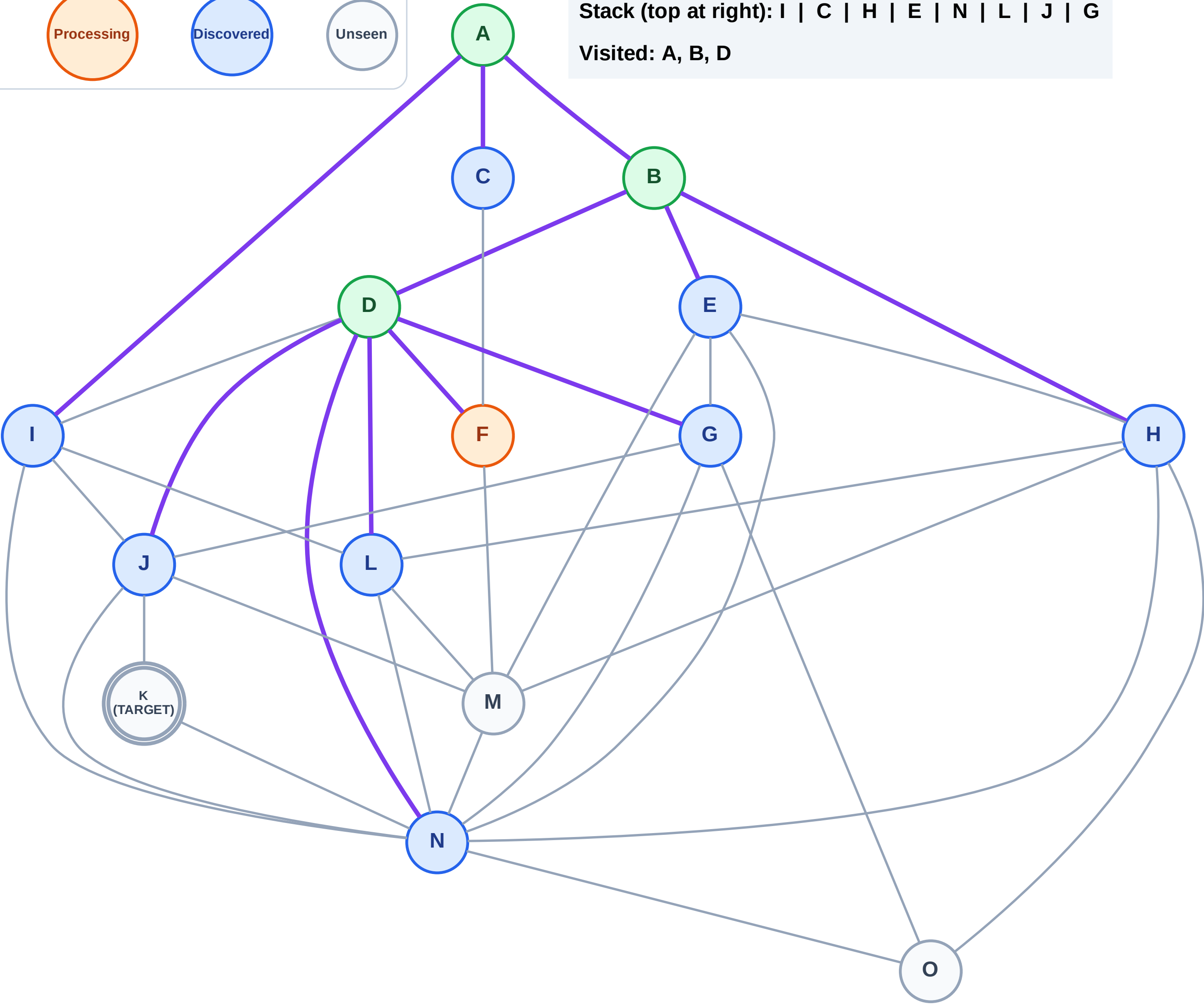
Processing

Discovered

Unseen

Stack (top at right): I | C | H | E | N | L | J | G

Visited: A, B, D



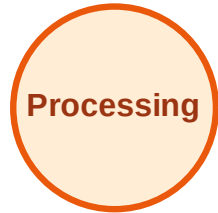
DFS Traversal

Step 9: Discover M from F

Legend



Complete



Processing



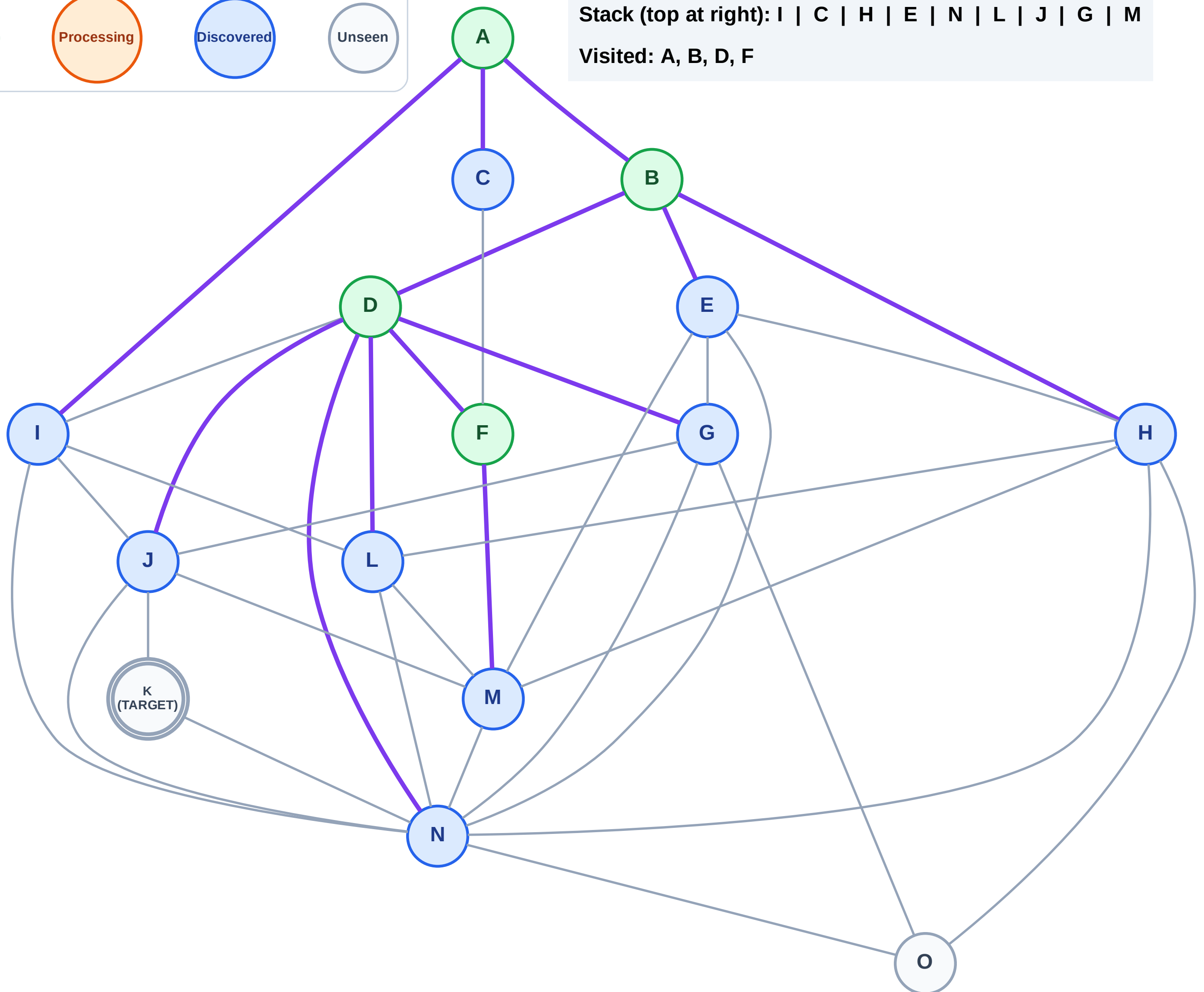
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L | J | G | M

Visited: A, B, D, F



DFS Traversal

Step 10: Process node M

Legend

Complete

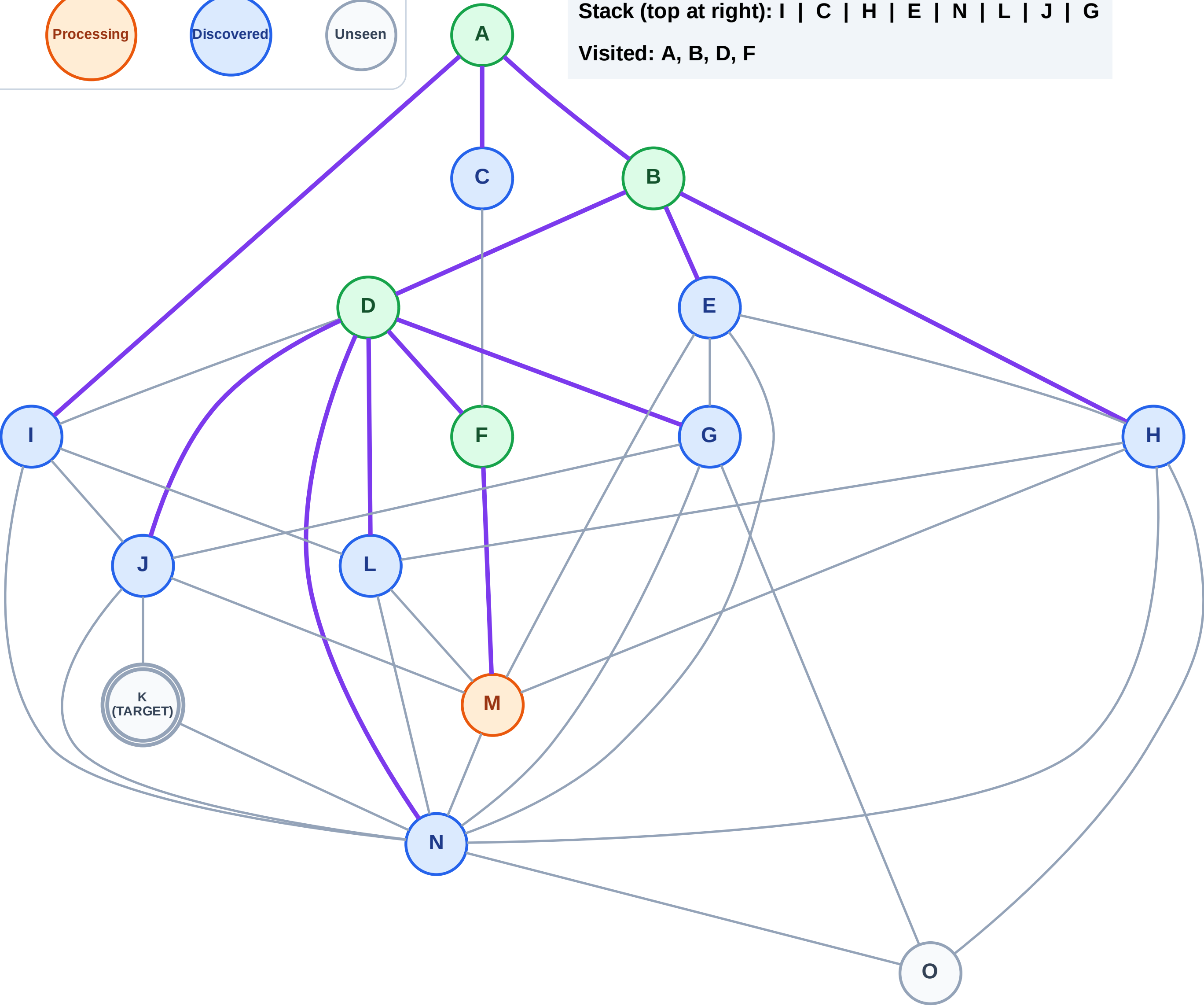
Processing

Discovered

Unseen

Stack (top at right): I | C | H | E | N | L | J | G

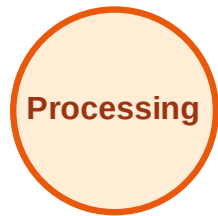
Visited: A, B, D, F



DFS Traversal

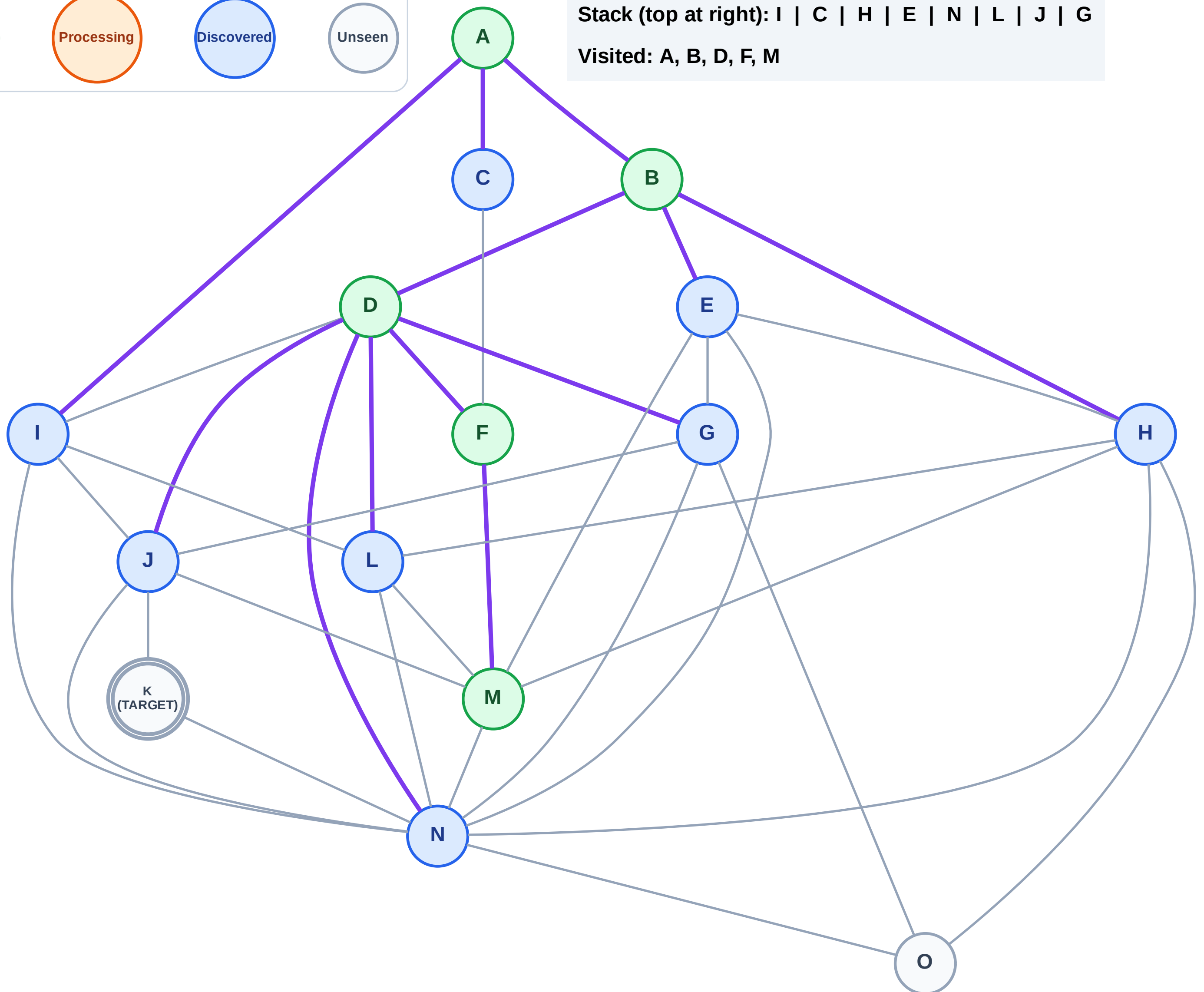
Step 11: No new neighbors from M

Legend



Stack (top at right): I | C | H | E | N | L | J | G

Visited: A, B, D, F, M



DFS Traversal

Step 12: Process node G

Legend

Complete

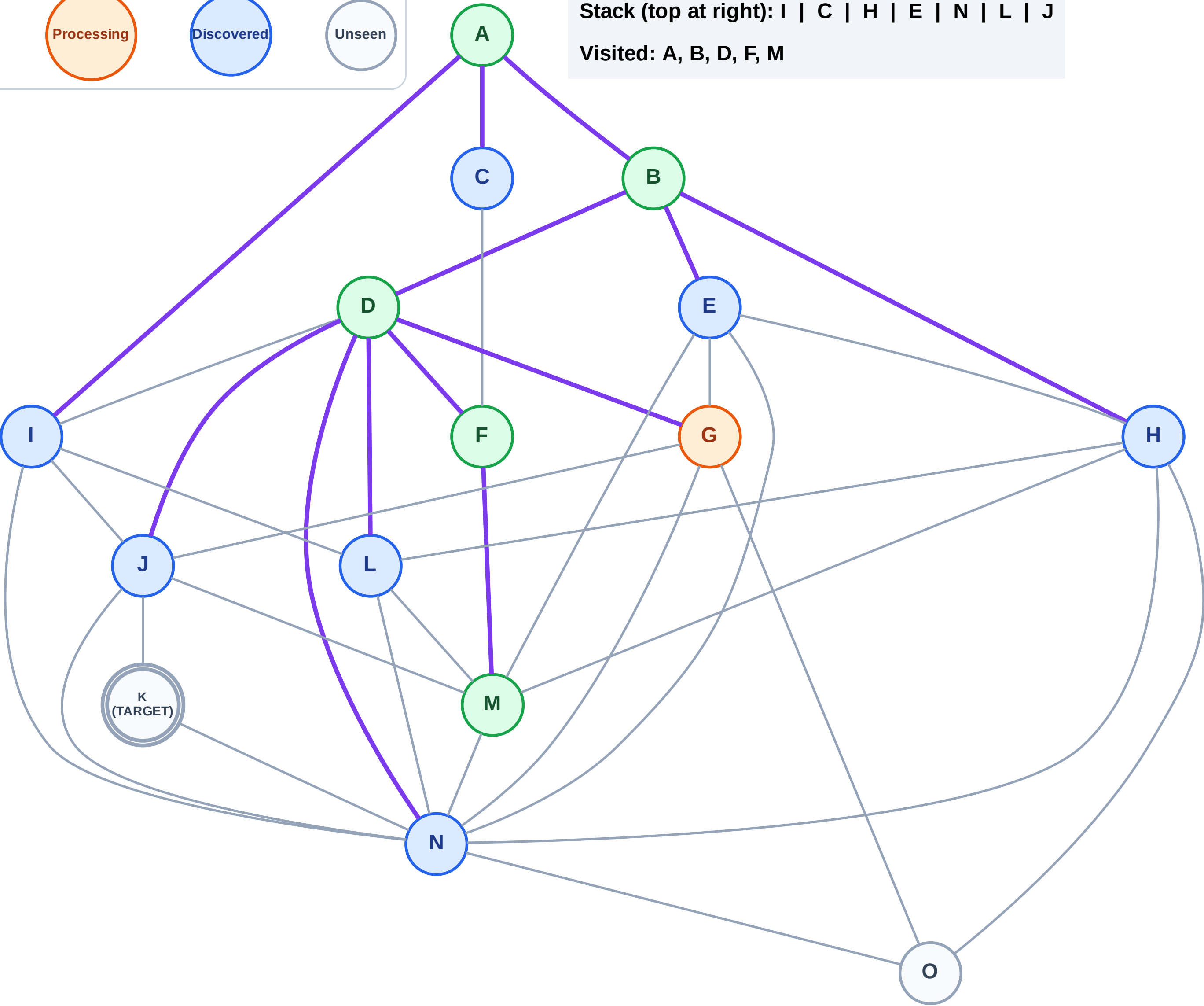
Processing

Discovered

Unseen

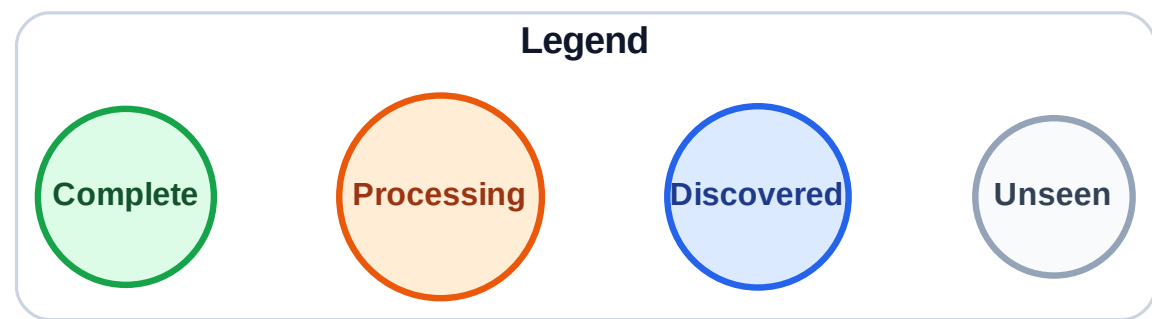
Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, M



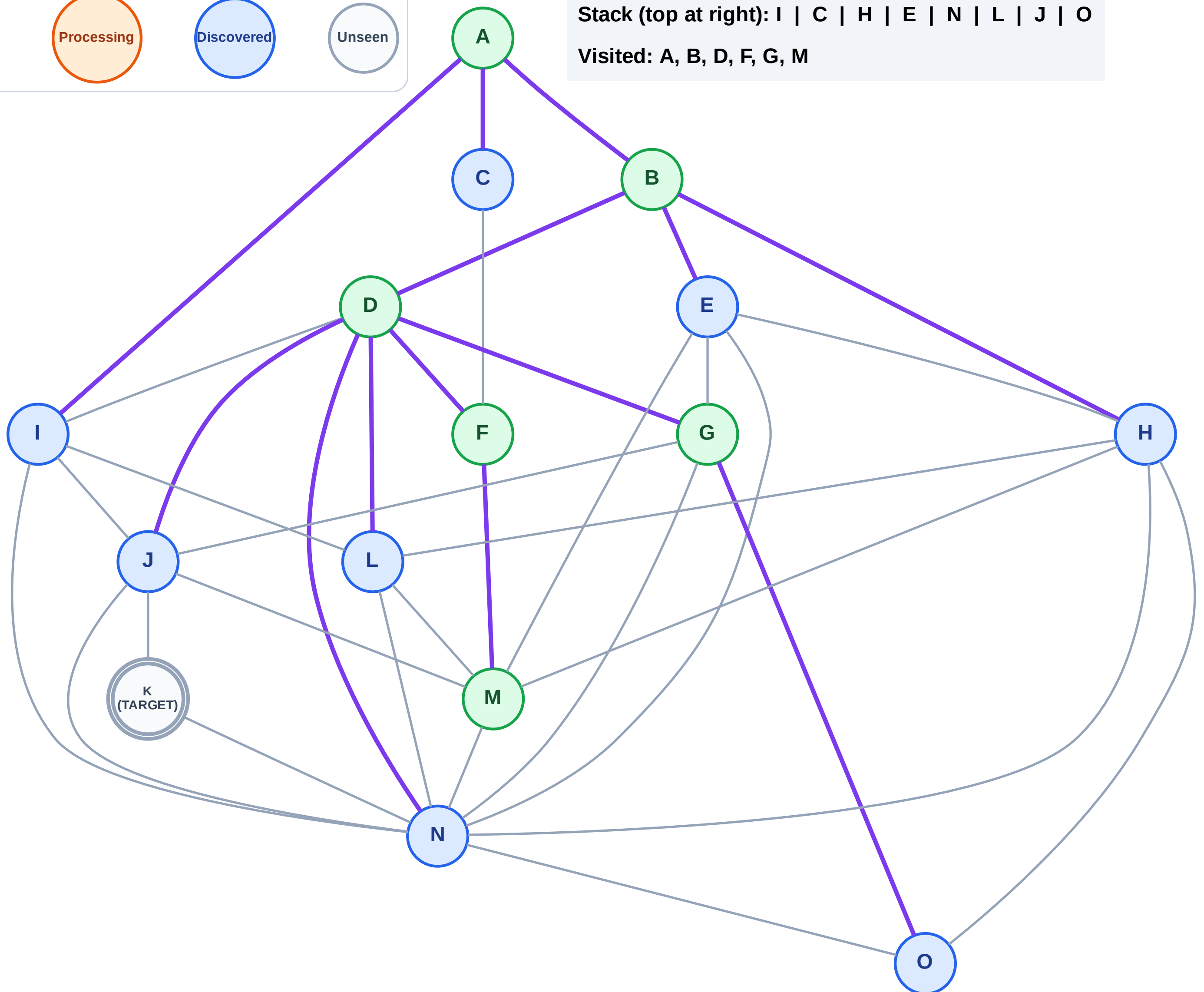
DFS Traversal

Step 13: Discover O from G



Stack (top at right): I | C | H | E | N | L | J | O

Visited: A, B, D, F, G, M



DFS Traversal

Step 14: Process node O

Legend

Complete

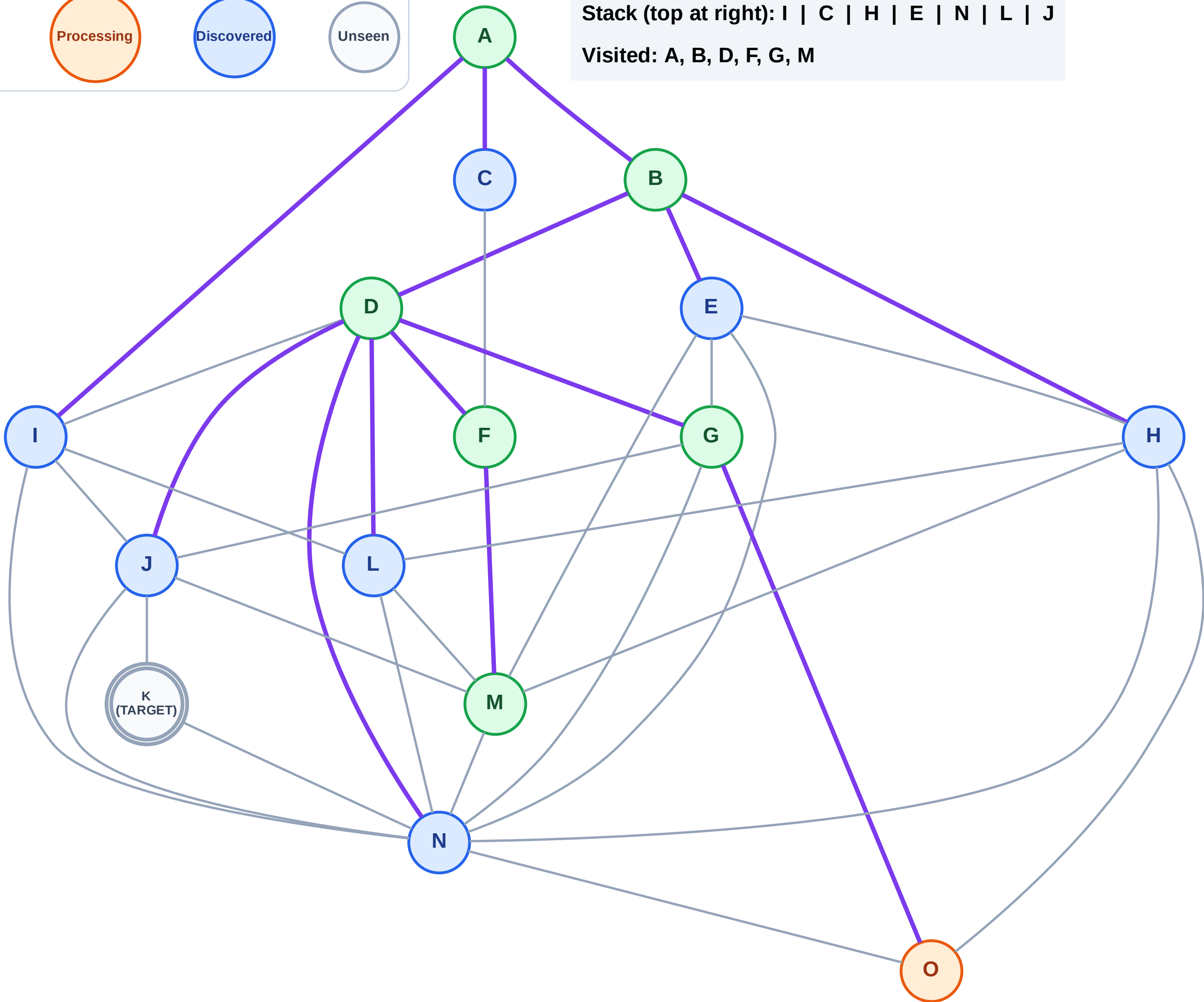
Processing

Discovered

Unseen

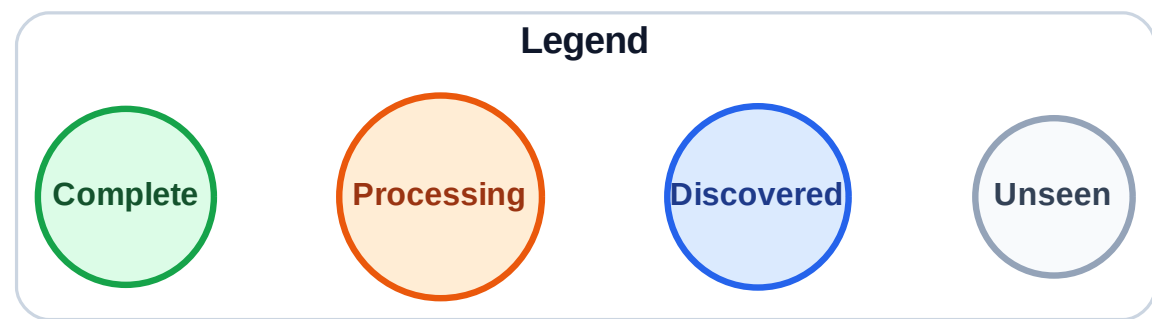
Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, G, M



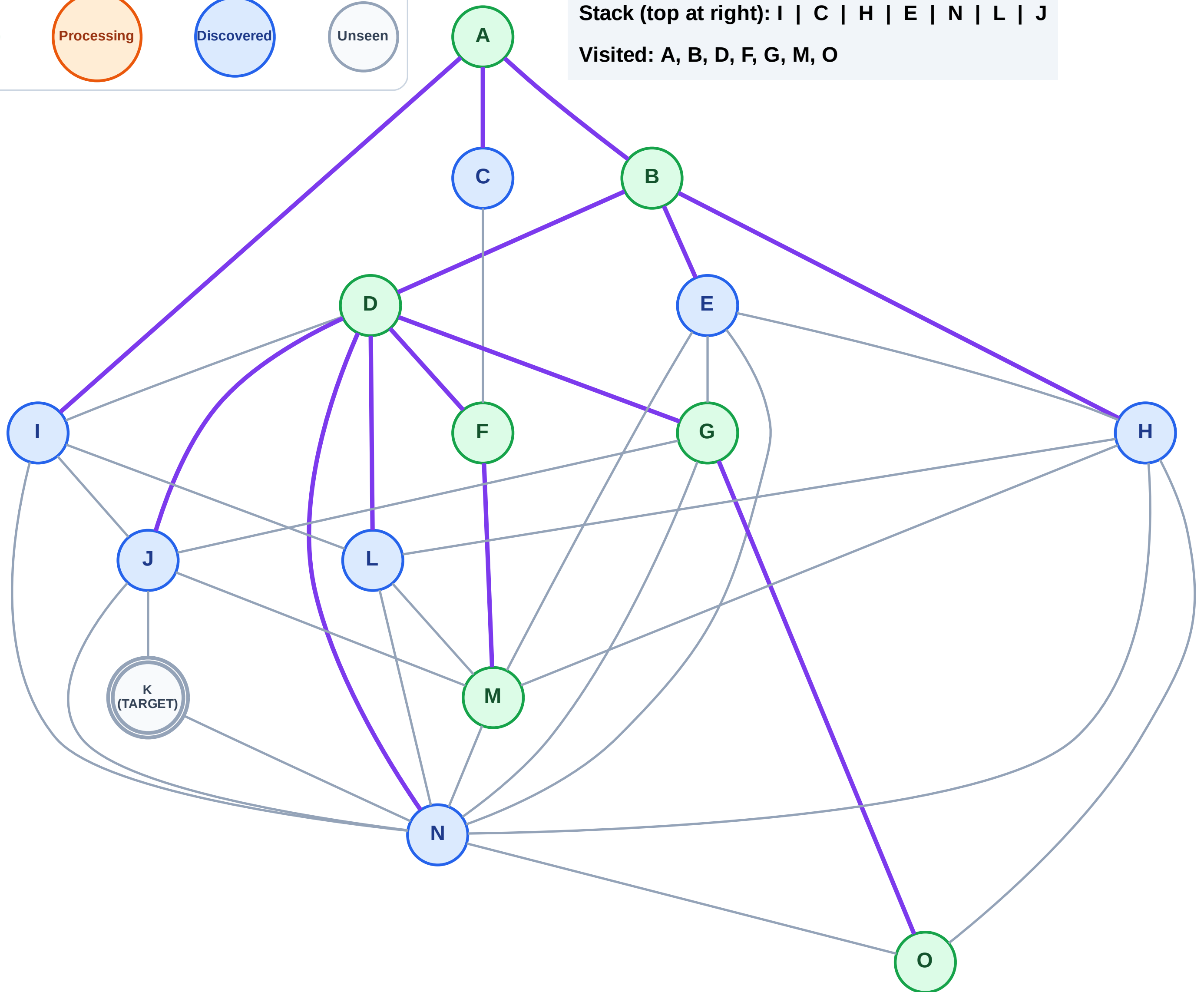
DFS Traversal

Step 15: No new neighbors from O



Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, G, M, O



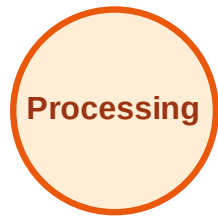
DFS Traversal

Step 16: Process node J

Legend



Complete



Processing



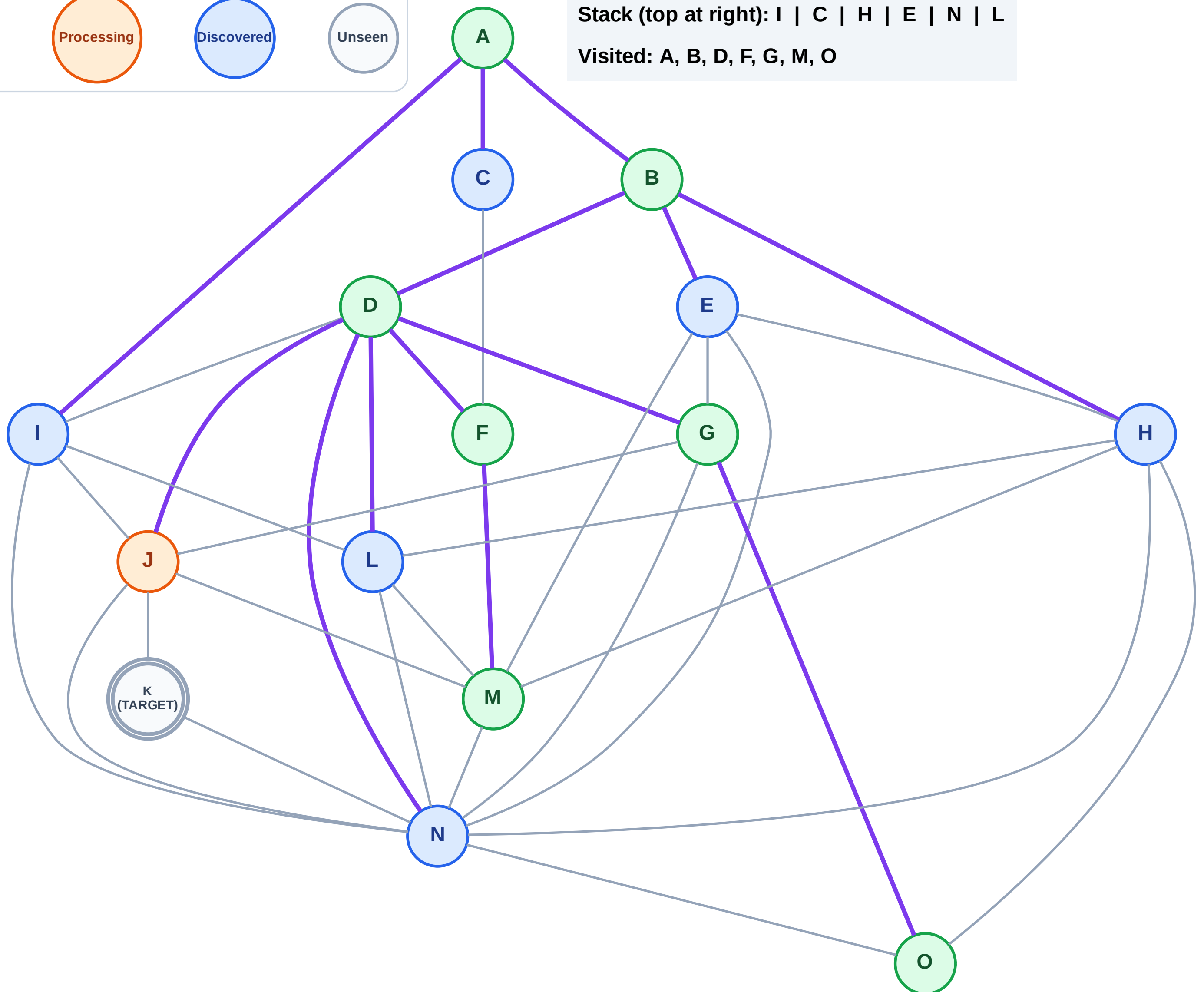
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L

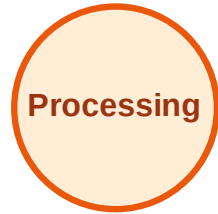
Visited: A, B, D, F, G, M, O



DFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → D → J → K

Legend



Stack (top at right): I | C | H | E | N | L

Visited: A, B, D, F, G, J, K, M, O

